NATIVE ANDROID | KOTLIN | JETPACK COMPOSE | NO WEBVIEW

WAQAR COMPUTER INSTITUTE

UI / UX Design & Motion Blueprint

Glassmorphism - Liquid Glass - Claymorphism - Neon Wave login
Admin, Teacher, Student and Parent portals: every screen, every function, every animation
Companion to the Native Android Architecture Blueprint v1.0 (450 functions)

Version 1.0 | 10 November 2026 | Status: proposed UI design for approval
Prepared for: Waqar (Owner / Administrator)

Design specification only. Kotlin source, Firebase deployment and device testing are separate delivery phases.

Document Control

Item	Detail
Document	UI / UX Design and Motion Blueprint - Waqar Computer Institute (native Android)
Version / date	1.0 / 10 November 2026
Companion document	Waqar Computer Institute - Native Android Architecture Blueprint v1.0 (450 functions, 4 portals)
Product type	Native Android app: Kotlin + Jetpack Compose. No WebView, no PWA, no web-app files of any kind.
Visual references	Attached Neon Wave login design (reference only); architecture diagrams from the blueprint
Audience	Owner, UI/UX designer, Android developers, QA
What is inside	Four materials (glass, liquid glass, clay, neon), design tokens, Compose component code, login portal, navigation shells, 24 admin modules with all 450 functions mapped to UI, three other portals, 32 animations, libraries, accessibility, tests, CI guards

How to read this document. Mockups are illustrations of the proposed design drawn for this document; they are not screenshots of a built app. Names and numbers in them are placeholders for explanation only and never ship in production.

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1. Purpose, Scope and Ground Rules

This document is the **UI / UX and motion specification** for the Waqar Computer Institute Android app. It builds directly on the Native Android Architecture Blueprint v1.0, which defined **450 functions**, **4 role portals**, a 5-tab admin bottom bar and a 7-group side menu. Here every one of those functions is given a concrete place on screen, a visual material (glass, liquid glass, clay or neon), and an animation.

The login look is based on the attached **Neon Wave login reference** (an HTML file). That file is used **only as a visual reference**. Nothing from it is shipped; every effect is rebuilt natively in Jetpack Compose (Chapter 5).

1.1 Non-negotiable rules

#	Rule	How this document enforces it
U1	Native Kotlin + Jetpack Compose only. No WebView, no PWA, no HTML/CSS/JS files in the app.	All effects are specified with Compose, AGSL shaders, Canvas, Filament or Lottie. Chapter 13 adds CI guards that fail the build if a WebView or web bundle appears.
U2	The attached HTML login is a reference, not source.	Chapter 5 maps every CSS / JS feature to a Compose equivalent and lists what must NOT be ported (demo login logic, Three.js CDN script, any-password success).
U3	Every one of the 450 functions has a UI home.	Chapter 7 and 8 list each function with its component, material and animation. Appendix A shows coverage per module.
U4	No dummy data in production.	All mockups in this PDF are illustrations. Names and numbers in them are not shipped; production screens show real empty states.
U5	Beautiful on every phone.	Four capability tiers (Ultra, High, Standard, Reduced) with a fallback for every effect (Chapter 4 and 12).
U6	No secrets in UI code.	No API keys in resources or assets. AI, Agora and payments go through Cloud Functions as in the architecture blueprint.

1.2 How to read the tables

Material codes used throughout: **GL** = glassmorphism, **LQ** = liquid glass, **CL** = claymorphism, **NE** = neon accent (glow, conic border, gradient CTA), **SYS** = system component (biometric prompt, share sheet). A code such as **GL+NE** means a glass surface with a neon accent.

Function IDs (F001 to F450) are the same IDs as Appendix A of the architecture blueprint. A "Type" of Screen, Action, List / filter, Report / export, Background job or Configuration keeps the meaning defined there.

> **Honest note.** The mockups are vector illustrations drawn to explain layout, material and motion. They are not screenshots of a built app. Exact pixel values will be tuned on real devices during the design-system phase; the tokens and rules here are the contract.

2. Design Language: Four Materials

The app combines four visual materials. Each has one job, so the interface stays calm and readable instead of becoming a pile of effects.

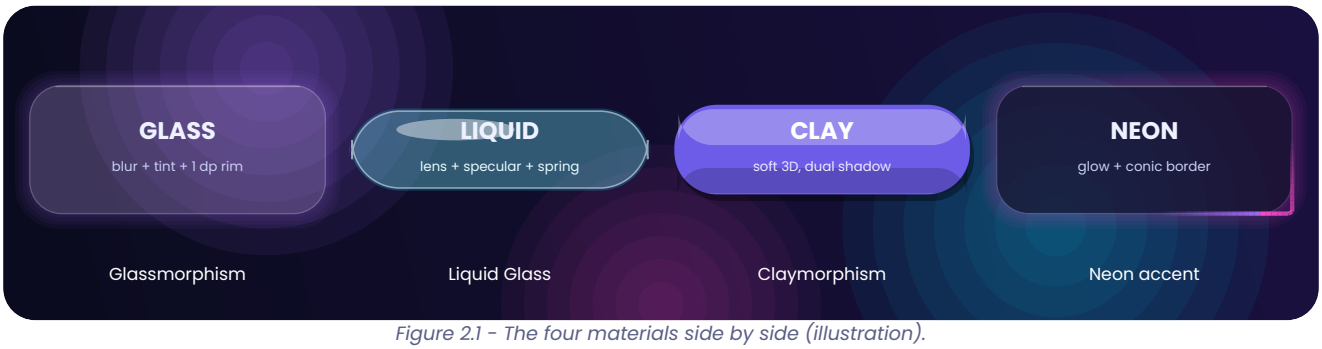


Figure 2.1 - The four materials side by side (illustration).

2.1 Material specifications

Material	What it looks like	Key values (starting point)	Used for
Glassmorphism (GL)	Frosted translucent panel over a colourful backdrop, thin bright rim, soft shadow.	Blur 16-28 dp; tint white 6-14% alpha; rim 1 dp white 22-30%; corner 16-28 dp; shadow 0/24 dp black 35%.	Cards, KPI tiles, list rows, input fields, bottom bar, drawer, charts.
Liquid glass (LQ)	Glass that looks like a drop of water: lens edge, bright specular highlight, shape that stretches and settles.	Blur 8-16 dp; tint white 12-20%; rim white 45-60%; specular ellipse top-left; spring damping 0.55-0.7.	Tab selection pill, chips, bottom sheets, dialogs, call controls over video, drag handles.
Claymorphism (CL)	Soft, inflated 3D buttons with light top-left and dark bottom-right shading; friendly and tactile.	Opaque saturated fill; two shadows (light -6/-6, dark 8/10); inner shade; radius 20-32 dp; press sinks 2 dp.	FAB, primary tiles on Student portal, grade stickers, certificate templates, destructive buttons, empty-state art.
Neon accent (NE)	Glowing gradient edge and call-to-action that pulls the eye to one thing per screen.	Glow shadow in role colour 35-55% alpha, 24-40 dp; conic border sweep 5 s; gradient 110 deg role A to B.	Primary CTA, focus ring, live / severity dots, login card, security posture tile.

2.2 Glass anatomy

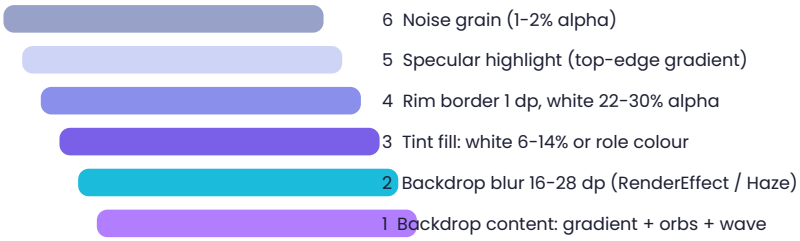


Figure 2.2 - The six layers of every glass surface. Layer 2 is the expensive one; Chapter 4 explains the fallbacks.

2.3 Which material goes where (policy)

UI element	Material	Rule
Screen background	Gradient + 2-3 soft colour orbs (+ wave on login/empty states)	Static or very slow drift. Never video. Orbs follow the role colour.
Cards, tiles, rows, charts	GL	Glass only when the number of glass items on screen is small; long lists use flat rows with a 1 dp divider.
Text input fields	GL (darker tint)	Floating label; focus = neon ring + 4 dp halo.
Bottom tab bar, drawer, top bar on scroll	GL	Bar blurs the content scrolling under it.
Tab selection indicator, chips, sheets, dialogs	LQ	Indicator stretches towards the target tab and settles with a spring.
Primary CTA (Login, Save, Post payment)	NE	Gradient fill, soft glow, shimmer sweep every 3.2 s while idle (stops when disabled or reduced motion).
FAB, status stickers, destructive actions, grade badges	CL	Clay never sits directly on a blurred layer; place it on a flat or gradient area.
Money and numbers	GL + tabular numerals	No effects on digits themselves; ticker animation only.
Video / live class controls	LQ over the video	Controls auto-hide after 3 s; never blur the video itself.

2.4 Portal personality

Portal	Accent pair	Dominant materials	Feel
Admin	#A970FF -> #FF4ECD (violet to pink)	GL + NE	Control centre: dense, precise, data-first.

Portal	Accent pair	Dominant materials	Feel
Teacher	#2DFF9B -> #D6FF3D (green to lime)	GL + LQ	Classroom: fast actions, big tap targets, minimal chrome.
Student	#00E1FF -> #4D7CFF (cyan to blue)	CL + GL	Learning hub: playful, rounded, encouraging.
Parent	#FFB02E -> #FF4D6D (amber to coral)	GL (soft) + CL	Family view: calm, reassuring, read-only clarity.

2.5 Do and do not

- **Do** use at most one neon CTA per screen and at most two live blur layers (Chapter 12).
- **Do** keep text contrast at 4.5:1 or better; add an 8% dark scrim behind text on bright backdrops.
- **Do** pair colour with an icon or label for status (attendance, dues) so colour-blind users are not excluded.
- **Do not** stack glass on glass more than one level deep.
- **Do not** animate layout properties on every frame inside long lists; animate graphicsLayer (alpha, scale, translation).
- **Do not** hard-code colours in screens; every colour comes from the theme tokens (a lint rule enforces this).

3. Design Tokens

Tokens are the single source of truth for colour, type, shape, spacing, depth and motion. They live in `:core:ui` and are the only way screens obtain a visual value.

3.1 Role colour pairs



Figure 3.1 - Accent pairs per portal, taken from the attached login reference.

3.2 Colour tokens

Token	Dark theme	Light theme	Use
background	#05060F (to #070816)	#F4F5FF	App background under gradients
surfaceGlass	rgba(14,18,42, 0.46)	rgba(255,255,255, 0.55)	Tint of glass cards and the login card
rim	rgba(255,255,255, 0.12-0.28)	rgba(255,255,255, 0.70)	1 dp border
textPrimary	#EEF1FF	#14162B	Headings and body
textMuted	#8D97C2	#5A5F85	Secondary text, labels
accentA / accentB	Role pair (3.1)	Role pair, 12% darker	CTA, focus, indicators, charts
success / present	#2DFF9B	#12A66A	Present, paid, healthy
warning / late / due	#FFB02E	#C27800	Late, due soon, quota near limit
danger / absent	#FF4D6D	#D6244A	Absent, overdue, destructive
info	#00E1FF	#0087A8	Neutral highlights, links

3.3 Typography

One variable sans family (recommended: a font with tabular numerals, bundled as a resource, SIL-OFL licensed). Money and counters use tabular numerals (`fontFeatureSettings = "tnum"`). Hindi strings use the system Devanagari font with the same scale. All sizes are in sp and support 200% font scaling.

Style	Size / line	Weight	Notes
Display	32 / 38	700	Login title, big KPI numbers (tabular)
Headline	24 / 30	700	Screen titles
Title	18 / 24	600	Card titles, sheet titles
Body	14 / 20	400	Default reading text
Label	11 / 16	600, +0.06 em	Chips, tab labels, tags such as CONTROL CENTER
Caption	10 / 14	400	Timestamps, helper text

3.4 Shape, spacing, depth

Token	Values	Notes
Radius	XS 8 - S 12 - M 16 - L 24 - XL 28 - Pill 50%	Login card = XL (28 dp, from the reference). Fields = M.
Spacing (4 dp grid)	4 - 8 - 12 - 16 - 20 - 24 - 32	Screen horizontal padding 16 dp (phones), 24 dp (tablets).
Touch target	Minimum 48 x 48 dp	Roster cells are 48 dp; compact tables are tablet-only.
Depth levels	E0 flat - E1 card - E2 sheet - E3 dialog	E1: shadow 12 dp 25% black; E3: 40 dp 55% black + role glow.
Blur radius	Thin 12 dp - Regular 20 dp - Thick 28 dp	Thick is for the drawer and sheets only.

3.5 Motion tokens

Token	Value	Source / use
durationInstant	90 ms	Colour change, ripple start
durationFast	150 ms	Chip select, icon swap
durationBase	250 ms	Banner, popup, fade-through
durationSlow	400 ms	Sheets, shared-axis, tab content
durationHero	550-1000 ms	Role content swap 550 ms; login card entrance 1000 ms (reference)
durationAmbient	3-5 s loop	Border sweep 5 s, CTA shimmer 3.2 s
easeStandard	cubic-bezier(.2, .9, .2, 1)	Reference "decelerate" feel for entrances and swaps
easeOvershoot	cubic-bezier(.5, 1.5, .5, 1)	Reference indicator motion; implemented as a spring in Compose
springSnappy	damping 0.7 / stiffness 600	Buttons, toggles, chips

Token	Value	Source / use
springBouncy	damping 0.55 / stiffness 300	Tab indicator, FAB, hand-raise
springGentle	damping 0.85 / stiffness 200	Drag settle, sheet snap

4. Building the Materials in Jetpack Compose

Each material is a reusable component or modifier in `:core:ui`. Screens never draw glass themselves; they call `GlassCard`, `LiquidPill`, `ClayButton` and `Modifier.neonBorder()`. This keeps fallbacks in one place.

> **Version note.** Library function names change between releases (for example the Haze blur API was renamed between 0.x and 1.x). The code below shows the pattern. Names are confirmed against the real library version when the design system is implemented.

4.1 Capability tier and reduce-motion

```
enum class UiTier { Ultra, High, Standard, Reduced }
val LocalUiTier = staticCompositionLocalOf { UiTier.Standard }

fun detectTier(ctx: Context): UiTier {
    val am = ctx.getSystemService<ActivityManager>()!!
    val pm = ctx.getSystemService<PowerManager>()!!
    val animScale = Settings.Global.getFloat(ctx.contentResolver, Settings.Global.ANIMATOR_DURATION_SCALE, 1f)
    val lowEnd = am.isLowRamDevice || am.memoryClass < 128
    return when {
        animScale == 0f || pm.isPowerSaveMode -> UiTier.Reduced      // user asked for less motion / battery saver
        lowEnd || Build.VERSION.SDK_INT < 31 -> UiTier.Standard      // no RenderEffect blur
        Build.VERSION.SDK_INT >= 33 && Build.VERSION.MEDIA_PERFORMANCE_CLASS >= Build.VERSION_CODES.S -> UiTier.Ultra
        else -> UiTier.High
    }   // a 2-second frame-time probe can demote the tier; Settings (F444) can override it
}
```

4.2 GlassCard (GL)

```
@Composable
fun GlassCard(
    modifier: Modifier = Modifier,
    glow: Color? = null,
    shape: Shape = RoundedCornerShape(24.dp),
    content: @Composable ColumnScope.() -> Unit,
) {
    val tier = LocalUiTier.current
    val g = LocalGlassTokens.current                    // tint, rim, blur radius from the theme
    val surface = when (tier) {
        UiTier.Ultra, UiTier.High ->                    // real backdrop blur (RenderEffect via Haze)
            Modifier.hazeEffect(state = LocalHazeState.current) { blurRadius = g.blur }
        else ->                                         // no blur: translucent gradient, still looks like glass
            Modifier.background(Brush.verticalGradient(listOf(g.tint.copy(.16f), g.tint.copy(.07f))))
    }
    Column(
        modifier
            .then(if (glow != null && tier != UiTier.Reduced) Modifier.glow(glow, 28.dp) else Modifier)
            .clip(shape).then(surface)
            .background(g.tint.copy(alpha = g.tintAlpha))
            .border(1.dp, Brush.verticalGradient(listOf(Color.White.copy(.35f), Color.White.copy(.10f))), shape) // rim + top
            .padding(16.dp),
        content = content,
    )
}
```

4.3 LiquidPill (LQ) behind an interface

Liquid glass is the most fragile part (libraries are young). The app therefore depends on a small interface, and the implementation can be swapped without touching any screen.

```
interface LiquidSurface {
    @Composable fun Pill(modifier: Modifier, selected: Boolean, accent: Color, content: @Composable RowScope.() -> Unit)
}
// Ultra : backdrop-capture library + AGSL lens shader (refraction, specular)      [wrapped, replaceable]
// High  : blur + specular highlight drawn with Canvas
// Std   : translucent gradient + 1 dp rim + static highlight
// Reduced: same as Std, no spring stretch

@Composable
fun SelectionPill(offset: Float /* 0..3 animated */, tabCount: Int, accent: Color, modifier: Modifier = Modifier) {
    BoxWithConstraints(modifier.fillMaxWidth().height(40.dp)) {
        val w = maxWidth / tabCount
        Box(Modifier.offset { IntOffset((offset * w.toPx()).roundToInt(), 0) }
            .width(w).fillMaxHeight().clip(RoundedCornerShape(50))
            .background(Brush.linearGradient(listOf(accent.copy(.85f), accent.copy(.55f))))
            .border(1.dp, Color.White.copy(.5f), RoundedCornerShape(50)))
    }
}
val pillSpring = spring<Float>(dampingRatio = 0.55f, stiffness = 300f)   // springBouncy
```

4.4 ClayButton (CL)

```
fun Modifier.clay(color: Color, radius: Dp = 24.dp, pressed: Boolean = false) = drawBehind {
    val r = radius.toPx(); val depth = (if (pressed) 2.dp else 8.dp).toPx()
    drawIntoCanvas { c ->
        fun shadowPass(dx: Float, dy: Float, sh: Color) {
            val p = Paint().asFrameworkPaint().apply {
                isAntiAlias = true; this.color = color.toArgb()
                setShadowLayer(depth * 1.6f, dx, dy, sh.toArgb())      // needs API 28+ for shapes
            }
            c.nativeCanvas.drawRoundRect(0f, 0f, size.width, size.height, r, r, p)
        }
        shadowPass(-depth, -depth, Color.White.copy(.28f))              // light, top-left
        shadowPass(depth, depth * 1.2f, Color.Black.copy(.40f))         // dark, bottom-right
    }
    drawRoundRect(color, cornerRadius = CornerRadius(r))                 // body
    drawRoundRect(Brush.verticalGradient(listOf(Color.White.copy(.30f), Color.Transparent), endY = size.height * .5f),
                  size = Size(size.width, size.height * .5f), cornerRadius = CornerRadius(r))   // top highlight
}
// Below API 28: replace the two shadow passes with two offset translucent rounded rects.
```

4.5 Neon border and shimmer CTA (NE)

```
fun Modifier.neonBorder(a: Color, b: Color, radius: Dp = 28.dp, enabled: Boolean = true) = composed {
    val t = rememberInfiniteTransition(label = "neon")
    val angle by t.animateFloat(0f, 360f, infiniteRepeatable(tween(5000, easing = LinearEasing)), label = "angle")
    drawWithContent {
        drawContent()
        if (!enabled) return@drawWithContent
        val sweep = android.graphics.SweepGradient(
            center.x, center.y,
            intArrayOf(0, 0, a.toArgb(), b.toArgb(), 0), floatArrayOf(0f, .6f, .7f, .85f, 1f)
        ).also { it.setLocalMatrix(android.graphics.Matrix().apply { postRotate(angle, center.x, center.y) }) }
        drawRoundRect(ShaderBrush(sweep), cornerRadius = CornerRadius(radius.toPx()), style = Stroke(1.5.dp.toPx()))
    }   // state read happens in the draw phase only, so there is no recomposition per frame
}

@Composable
fun ShimmerButton(text: String, a: Color, b: Color, enabled: Boolean, onClick: () -> Unit) {
    val x by rememberInfiniteTransition(label = "sh").animateFloat(-1.2f, 1.2f,
        infiniteRepeatable(tween(3200, easing = LinearEasing)), label = "x")
    Box(Modifier.fillMaxWidth().height(54.dp).clip(RoundedCornerShape(16.dp))
        .background(Brush.linearGradient(listOf(a, b))).clickable(enabled = enabled, onClick = onClick)
        .drawWithContent { drawContent()
            if (enabled) drawRect(Brush.horizontalGradient(listOf(Color.Transparent, Color.White.copy(.55f), Color.Transparen
                startX = size.width * x, endX = size.width * (x + .4f))) },
        contentAlignment = Alignment.Center) { Text(text, fontWeight = FontWeight.ExtraBold, color = Color(0xFF05060F)) }
}
```

4.6 Performance rules for materials

- At most **two** live blur layers on screen; blur only the region of the card, never the full screen.
- One shared backdrop (`HazeState`) per screen; do not create a state per row.
- Animated values are read in the draw or graphicsLayer phase (no per-frame recomposition).
- Lists over about 30 visible rows use flat rows with a 1 dp divider; glass is applied to headers and sheets only.
- The 3D hero (Filament) and the wave shader pause when the screen is not visible or the device is hot (thermal status moderate or above).

5. Login Portal: Neon Wave

The attached login file defines one card with four role tabs (Admin, Student, Teacher, Parent). Changing the tab changes the accent colours, the heading, the field label and the button text, and sends a ripple through a particle wave in the background. This chapter keeps that design and rebuilds it natively.

5.1 What the attached design contains

Element	Reference behaviour	Native equivalent
Background	Animated 3D particle wave (WebGL, 150 x 150 points on desktop, 90 x 90 on mobile) that reacts to the pointer, with a CSS gradient fallback.	AGSL shader (API 33+) or Canvas points (older), static gradient on Reduced tier.
Brand mark	Gradient square with a letter and a two-line wordmark, top-left.	Compose Box + Text; brand name and colours come from `InstituteBrand` config.
Card	28 px radius, glass fill rgba(14,18,42,.46), blur 22 px, 1 px rim, big black shadow, role-colour glow.	GlassCard (XL radius) + glow + neonBorder.
Border sweep	Conic gradient ring rotating every 5 s.	Modifier.neonBorder (4.5).
Role tabs	Four tabs with icons and labels; gradient indicator slides with an overshooting curve (550 ms).	SelectionPill driven by a bouncy spring (4.3).
Role content	Tag, title, subtitle, ID label and button text swap with a blur + fade + rise (550 ms).	AnimatedContent with fade, slide 14 dp, scale .98, optional blur on API 31+.
Fields	Floating labels, 56 px height, neon focus ring and halo, show/hide eye icon.	OutlinedTextField replacement with animated label; eye toggles visual transformation.
CTA	Gradient button with light sweep every 3.2 s; spinner while verifying; shake on error.	ShimmerButton (4.5), CircularProgressIndicator, Animatable shake.
Success	Wave pulse ring plus raised wave amplitude.	Canvas expanding rings + shader amplitude uniform.
Tilt	Card rotates up to 7 degrees following the mouse (disabled on touch).	Gyroscope tilt with graphicsLayer rotation, same 7 degree cap; off on Reduced tier.
Chips row	Encrypted, Secure, 24/7 with glowing dots.	Row of small labels with glow dots.
Reduced motion	Sweep and shimmer stop; wave calmer.	Reduced tier: static gradient, no sweep, no shimmer, cross-fade only.

5.2 The four role screens



Figure 5.1 - Login for Admin, Student, Teacher and Parent. Accent colours crossfade over 700 ms when the tab changes.

5.3 Login states

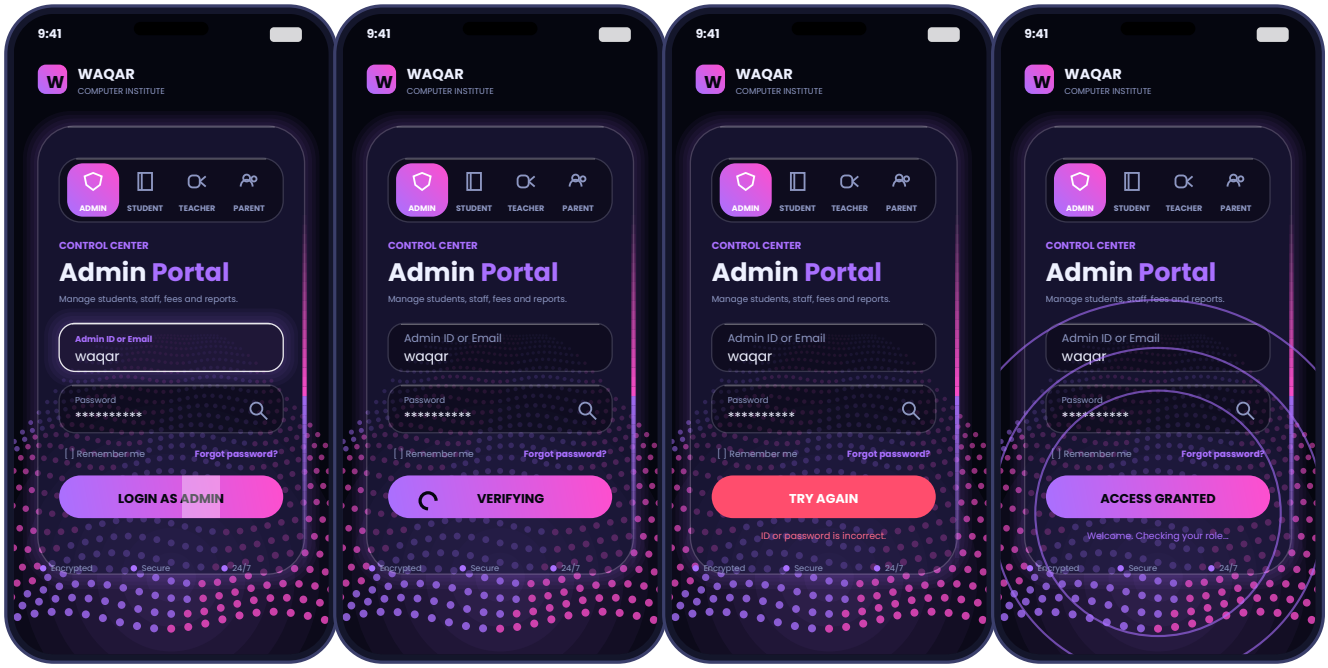


Figure 5.2 - Focus, verifying, wrong password and access granted.

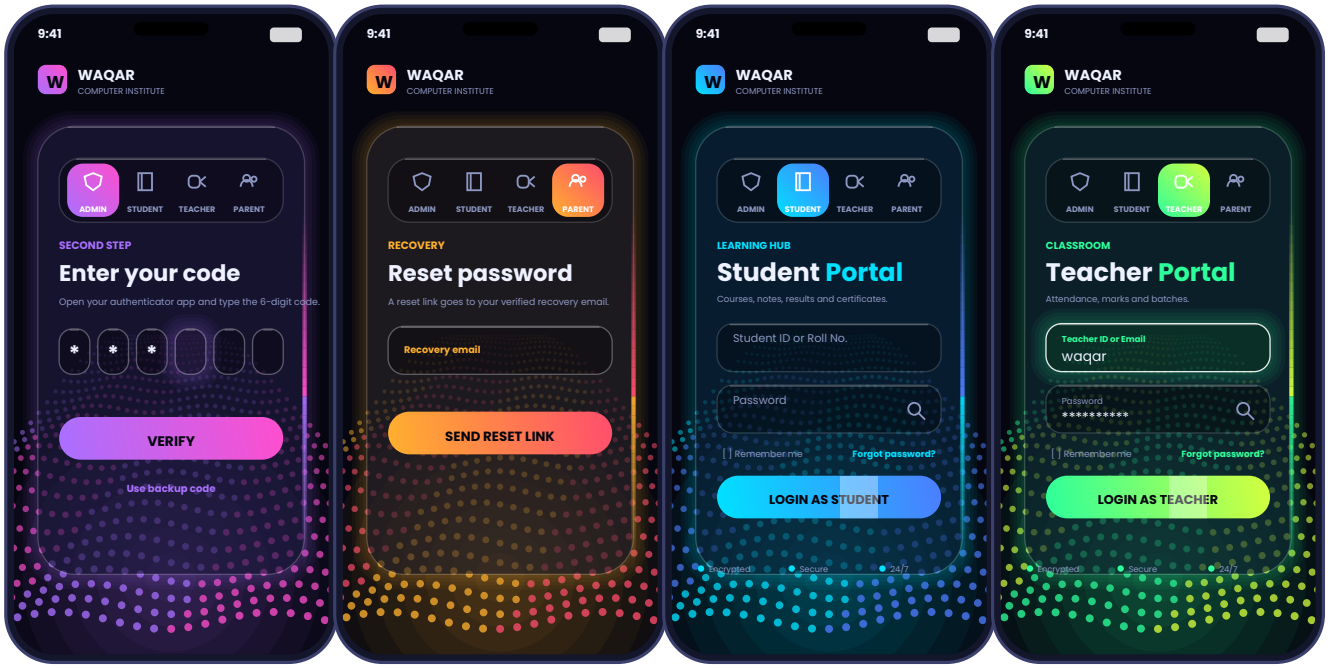


Figure 5.3 - Second factor step (admin), password reset, student idle, teacher focus.

State	Visual	Motion	Real behaviour (from the architecture)
Idle	Fields empty, shimmer running	Card entrance 1000 ms on first show	Role tab is a hint only
Focus	Neon ring + 4 dp halo, label floats up	220 ms label move	-
Validating	Inline red hint under field	Field shake 420 ms	Lowercase username, length checks (F003)
Verifying	Spinner + text VERIFYING, button disabled	Spinner 700 ms per turn	signInWithEmailAndPassword(alias, password) (F002)
Second factor	6-box code input	Box focus glow, auto-advance	Admin MFA mandatory; TOTP (F010)
Success	ACCESS GRANTED, pulse rings, wave rises	1.5 s ring + 600 ms route transition	Claims decide the portal; navigate to its home
Wrong password	Red CTA, message, shake	Shake -9 / 9 / -5 / 0 px over 420 ms	Counter increases; progressive delay (C08)
Locked	Amber banner with timer	Countdown ticker	Lockout after repeated failures
Offline	Info chip: "No internet"	Chip slide-down	Sign-in needs network; cached session can open app
Forgot password	Liquid sheet with recovery-email field	Sheet slide-up 400 ms	Reset mail goes to the verified recovery email (F006)
Biometric resume	System prompt over blurred app	Blur 200 ms	App lock after background timeout (F004)

State	Visual	Motion	Real behaviour (from the architecture)
Session expired	Banner on login	Slide-down 250 ms	F018, F015

5.4 What must not be copied from the HTML file

- **Demo sign-in.** The reference accepts any ID with a 4-letter password and prints "Demo". The app signs in only through Firebase Authentication.
- **Three.js script from a CDN.** No web script of any kind is allowed. The wave is drawn with AGSL / Canvas.
- **Mixed Hindi-English copy inside the layout.** All text goes to string resources with English and Hindi translations (the architecture lists Hindi as an owner decision).
- **Role tab as security.** Choosing "Admin" in the UI grants nothing. The server-set role claim decides the portal. If the claim differs from the tab, the app switches the tab and shows a neutral message.
- **Parent login.** The blueprint uses phone OTP for guardians; the Parent tab therefore shows a mobile-number step followed by the 6-box code. Password-based guardian login is an open owner decision.
- **Brand name.** The reference carries a different institute name. The wordmark and mark are configuration so the Waqar Computer Institute name and logo are used.

5.5 Animation timeline of one login attempt

Time	Event	Animation
0 ms	Screen opens	Card rises 40 dp, rotates from 18 degrees, scales .96 to 1 (1000 ms, easeStandard); wave fades in 600 ms
0–5000 ms loop	Idle	Conic border sweeps once every 5 s; CTA shimmer every 3.2 s; wave drifts
Tap role tab	Role change	Pill springs to new tab (bouncy); accent colours crossfade 700 ms; content swaps in 550 ms; wave ring pulse 1.5 s
Tap LOGIN	Verify	Button text swaps to spinner + VERIFYING; fields disabled
+0.4 to 1.5 s	Server answers	Success: ring pulse + wave boost, tick, route fade. Error: shake 420 ms, red CTA for 2.2 s then back

5.6 Role tabs and the screen skeleton

```
data class RolePalette(val role: Role, val a: Color, val b: Color, val tag: String, val titleRes: Int, val idLabelRes: Int, v

@Composable
fun LoginScreen(vm: LoginViewModel = hiltViewModel()) {
    val ui by vm.state.collectAsStateWithLifecycle()
    val roles = remember { RolePalettes.all }                        // Admin, Student, Teacher, Parent
    var sel by rememberSaveable { mutableIntStateOf(0) }
    val a by animateColorAsState(roles[sel].a, tween(700), label = "a")
    val b by animateColorAsState(roles[sel].b, tween(700), label = "b")
    val shift by animateFloatAsState(sel.toFloat(), spring(dampingRatio = 0.55f, stiffness = 300f), label = "pill")
    val shake = remember { Animatable(0f) }
    LaunchedEffect(ui.errorTick) { if (ui.errorTick > 0) shake.animateTo(0f, keyframes {
        durationMillis = 420; -9f at 105; 9f at 210; -5f at 315; 0f at 420 }) }

    Box(Modifier.fillMaxSize().background(Color(0xFF05060F))) {
        NeonWave(a, b, pulse = ui.pulseTick, boost = ui.success)             // 5.7
        BrandMark(Modifier.align(Alignment.TopStart).statusBarsPadding().padding(22.dp, 18.dp))
        GlassCard(Modifier.align(Alignment.Center).widthIn(max = 420.dp).padding(14.dp)
                .offset { IntOffset(shake.value.dp.roundToPx(), 0) }
                .tiltWithGyro(maxDeg = 7f).neonBorder(a, b), glow = a, shape = RoundedCornerShape(28.dp)) {
            RoleTabs(shift, roles, sel, onSelect = { sel = it; vm.onRoleHint(roles[it].role) })
            AnimatedContent(sel, transitionSpec = {
                (fadeIn(tween(550, easing = CubicBezierEasing(.2f, .9f, .2f, 1f))) + slideInVertically(tween(550)) { 14.dp.ro
            }, label = "role") { i -> RoleForm(roles[i], ui, onSubmit = vm::signIn) }
        }
    }
}
```

5.7 The wave background by tier

Tier	Technique	Cost control
Ultra (API 33+)	AGSL RuntimeShader paints a perspective dot grid in one draw call; uniforms: time, resolution, two colours, ring radius, boost.	Pause when not resumed; render at 0.75 scale on thermal moderate.
High / Standard	Canvas drawPoints with a pre-allocated FloatArray; about 40 x 30 points, updated at 30 fps.	Fewer points on low RAM; no per-frame allocation.
Reduced	Static gradient + orbs, no motion.	Zero GPU work.

```
const val WAVE_AGSL = """
uniform float2 iResolution; uniform float iTime;
layout(color) uniform half4 colorA; layout(color) uniform half4 colorB;
half4 main(float2 fc) {
    float2 uv = fc / iResolution;
    float horizon = 0.38;
    if (uv.y < horizon) return half4(0.0);
    float z = 1.0 / (uv.y - horizon + 0.05);                      // perspective depth
    float2 g = float2((uv.x - 0.5) * z * 6.0, z * 2.0 + iTime * 0.6);
    float w = sin(g.x * 0.9 + iTime) + cos(g.y * 0.8 - iTime * 0.7);   // wave height
    float2 cell = fract(g) - 0.5;
    float d = length(cell + float2(0.0, w * 0.12));
    float dotMask = smoothstep(0.18, 0.0, d);
    float fade = smoothstep(0.0, 0.5, uv.y - horizon);
    half3 col = mix(colorA.rgb, colorB.rgb, half(clamp(w * 0.25 + 0.5, 0.0, 1.0)));
    return half4(col * dotMask * fade, dotMask * fade);
}"""   // reference shader: look and speed are tuned on real devices
```

5.8 Wiring to real sign-in

- `LoginViewModel.signIn()` validates input, lowercases the username, builds the internal alias, calls the sign-in use case (architecture chapter 6.6) and emits `LoginUiState`: Idle, Verifying, NeedsMfa, Success(role), Error(kind).
- On Success the app reads token claims and routes to the matching portal shell (Chapter 6).
- App Check token is attached automatically; no credential is stored on the device; biometrics only re-open an existing session.

6. Navigation Shells and Transitions

After sign-in the app resolves the role claim and loads one **shell**: a top bar, a body, and a navigation system. The admin shell has two navigation systems that never duplicate each other: the five-tab bottom bar for daily work and a side drawer for everything else.



Figure 6.1 - Admin shell: Home tab, side drawer (Batches selected), Students tab, Inbox tab.

#	Element	Material	Behaviour and motion
1	Bottom tab bar (5 tabs)	GL with LQ selection pill	Pill slides and stretches with the bouncy spring; icon scales 1.0 to 1.15; label fades in. Tab bar blurs content scrolling underneath. Hides on scroll down in long lists, returns on scroll up.
2	Side drawer (7 groups)	GL (thick blur)	Opens by edge swipe or menu icon; content behind scales to 0.94 and dims; items stagger in 30 ms each. Selected item uses a liquid pill.
3	Top app bar	GL on scroll	Transparent at top; becomes glass when content scrolls under it.
4	Profile menu (avatar, top right)	LQ popup	Password, username, sessions, sign out (module M01). Opens with scale-from-anchor 180 ms.
5	Tablets (width 600 dp+)	GL	Drawer becomes a permanent navigation rail or panel; two-pane list-detail (Students list + profile).

6.1 Tabs and menus per portal

Portal	Bottom tabs (5)	Side menu
Admin	Home, Students, Attendance, Fees, Inbox	People (Admissions, Staff, Guardians); Academics (Courses, Batches, Exams, Material, Certificates, Live Classes, AI Hub); Finance (Accounts, Payroll); Insights (Reports); Operations (Inventory, Support); System (Security, Backup, Settings)
Teacher	Today, Attendance, Batches, Material, Inbox	Exams and marks, Live classes, AI tutor, Leave and payslip, Support, Profile
Student	Home, Timetable, Fees, Results, Inbox	Attendance, Material and assignments, Live classes, AI tutor, Certificates, Support, Profile
Parent	Children, Attendance, Fees, Results, Inbox	Timetable, Notices, Message teacher, Consent and preferences, Support, Profile

6.2 Transition rules

Transition	Pattern	Duration	Compose API
Tab to tab	Fade-through (no horizontal travel)	250 ms	AnimatedContent + fadeIn/fadeOut
Drawer item to screen	Shared axis X (slide 30 dp + fade)	300 ms	Navigation Compose enter/exit transitions
List row to detail	Container transform (card grows into screen)	400 ms	SharedTransitionLayout + sharedBounds
Open bottom sheet	Slide up, backdrop dims to 40%	400 ms spring gentle	ModalBottomSheet with custom glass container
Back	Predictive back preview (Android 14+), fallback to reverse of enter	System	PredictiveBackHandler / Navigation back animations
Login to portal	Success rings, card scales out, shell fades in	600 ms	Crossfade between login and shell graph

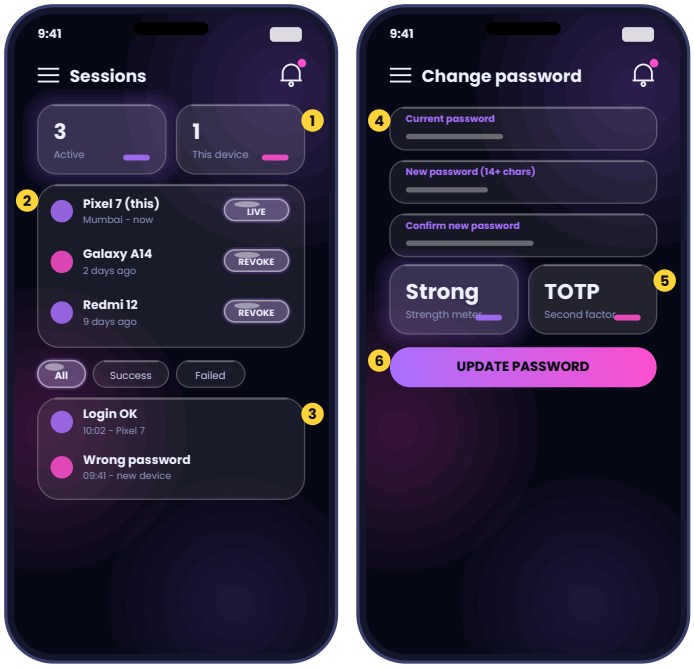
7. Admin Portal: Every Screen and Function

Each module below has two annotated screens, the material and motion summary, and a **complete function table**: every function ID from the architecture blueprint with the screen that hosts it, the UI component that implements it, the material and the animation. Yellow numbered dots on the mockups match the numbered notes beside them.

> **Reading the function tables.** "[screen]" marks a full destination reachable from the tab bar or drawer. Other rows live inside that screen as buttons, sheets, chips, fields or background chips. Job rows have no screen; they appear only as a status chip or notification.

M01 Authentication & Session (18 functions)

Navigation: Pre-login flow + Profile menu (avatar) **Roles:** All roles



Materials: Glass fields and rows, neon focus ring, clay danger button, liquid sheet for QR.

Motion: Swipe reveal, field focus glow, shake on error, ticker on strength meter.

1 Glass counter tiles show active sessions and the current device; the current device carries a neon LIVE chip (F011, F018).

2 Each session row supports swipe-to-revoke; the red clay REVOKE button reveals under the glass row (F012, F013).

3 Login attempt history (F017) filters with liquid chips; failed rows pulse the severity dot once.

4 Fields use floating labels; focus draws a neon ring (F007).

5 Strength meter and TOTP tile are glass; TOTP enrolment (F009) opens a liquid bottom sheet with a QR.

6 Primary CTA is gradient neon with a shimmer sweep; wrong re-authentication triggers a 420 ms shake.

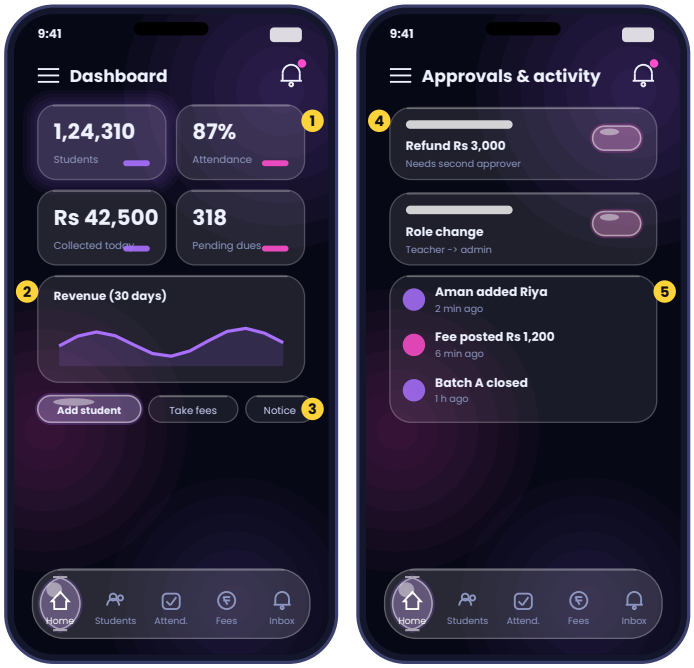
Figure 7.1 - Authentication & Session: Sessions (left) and Change password (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F001	Login screen (username + password)	Screen	**[screen]** Pre-login flow + Profile menu (top-right avatar)	Neon Wave login card (see Chapter 5)	GL+NE	Entrance 1000 ms, conic border sweep 5 s
F002	Sign in with username alias	Action	Login screen (username + password)	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms
F003	Input validation and lockout counter	Action	Login screen (username + password)	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F004	Biometric unlock on app resume	Action	Login screen (username + password)	System BiometricPrompt; app-lock blur overlay	SYS	Overlay blur 200 ms, unlock fade
F005	Remember trusted device	Action	Login screen (username + password)	Session row with swipe-to-revoke	GL+CL	Swipe reveal red clay button, row collapse 200 ms
F006	Forgot password via verified email	Action	Login screen (username + password)	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms
F007	Change password (re-authentication required)	Action	Login screen (username + password)	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms
F008	Change username (server callable)	Action	Login screen (username + password)	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms
F009	Enrol second factor (TOTP)	Action	Login screen (username + password)	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms

ID	Function	Type	Placed in	UI component	Mat	Animation
F010	Verify second-factor challenge	Action	Login screen (username + password)	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F011	Active sessions list	Screen	**[screen]** Pre-login flow + Profile menu (top-right avatar)	Session row with swipe-to-revoke	GL+CL	Swipe reveal red clay button, row collapse 200 ms
F012	Revoke single session	Action	Active sessions list	Session row with swipe-to-revoke	GL+CL	Swipe reveal red clay button, row collapse 200 ms
F013	Sign out all devices	Action	Active sessions list	Session row with swipe-to-revoke	GL+CL	Swipe reveal red clay button, row collapse 200 ms
F014	Auto-logout on idle	Job	Active sessions list	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F015	Token refresh and claim sync	Job	Active sessions list	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F016	Force re-login after role change	Job	Active sessions list	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F017	Login attempt history	List	Active sessions list	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F018	Session expiry banner	Action	Active sessions list	Session row with swipe-to-revoke	GL+CL	Swipe reveal red clay button, row collapse 200 ms

M02 Admin Dashboard (16 functions)

Navigation: TAB 1: Home **Roles:** Admin



Materials: Glass tiles + neon glow, liquid pills, parallax tilt on tiles on Ultra tier.

Motion: Count-up, chart draw-in, parallax tilt (gyro), list slide-in, pull-to-refresh liquid drop.

1 KPI tiles (F020-F026) are glass with a neon glow on the first tile; values count up in 600 ms and tap opens the matching report.

2 Charts (F031, F032) are Vico inside a glass card; the line draws in 700 ms and the tooltip follows the finger with a spring.

3 Quick-actions bar (F028) is a row of liquid pills; long-press reorders them (F034).

4 Pending approvals queue (F029) uses glass cards with Approve / Reject; money items demand re-authentication.

5 Recent activity feed (F030) is the audit tail; new rows slide in from the top and highlight for 1.2 s.

Figure 7.2 - Admin Dashboard: Dashboard (left) and Approvals & activity (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F019	Home dashboard	Screen	**[screen]** TAB 1: Home	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring
F020	Student count tile (aggregate query)	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97
F021	New admissions this month tile	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97
F022	Today's attendance tile	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97
F023	Fee collected today tile	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97
F024	Pending dues tile	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97
F025	Upcoming batches tile	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97
F026	Upcoming exams tile	List	Home dashboard	KPI glass tile (tap = drill-down)	GL+NE	Count-up 600 ms, press scale .97

ID	Function	Type	Placed in	UI component	Mat	Animation
F027	Birthdays and anniversaries card	List	Home dashboard	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F028	Quick actions bar	Action	Home dashboard	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F029	Pending approvals queue	List	Home dashboard	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F030	Recent activity feed (audit tail)	List	Home dashboard	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F031	Revenue trend chart	Report	Home dashboard	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F032	Enrolment trend chart	Report	Home dashboard	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F033	Pull-to-refresh and live-listener toggle	Action	Home dashboard	Settings switch / slider row	GL	Thumb spring, haptic confirm
F034	Dashboard widget layout preference	Config	Home dashboard	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring

M03 Student Management (30 functions)

Navigation: TAB 2: Students **Roles:** Admin (write); Teacher (read, own batches)



Materials: Glass rows and fields, liquid filter pills, clay status badges.

Motion: Stagger list, chip morph, upload ring, button-to-tick morph.

1 Search (F036, F037) is a glass field with 300 ms debounce; prefix search runs on nameLower.

2 Filter chips (F038-F041) are liquid pills; the selected chip morphs into the sheet header when "Filter" opens.

3 Paged list (F035) uses Paging 3 with 25-50 rows; glass is applied to rows only because the page is small. Shimmer placeholders fill while loading.

4 Photo and ID tiles (F046, F047) open a media sheet with CameraX and a circular upload ring.

5 Duplicate phone check (F044) runs on blur and shows an inline glass warning chip.

6 SAVE (F043) is the neon CTA; success morphs the button to a tick and the list scrolls to the new row.

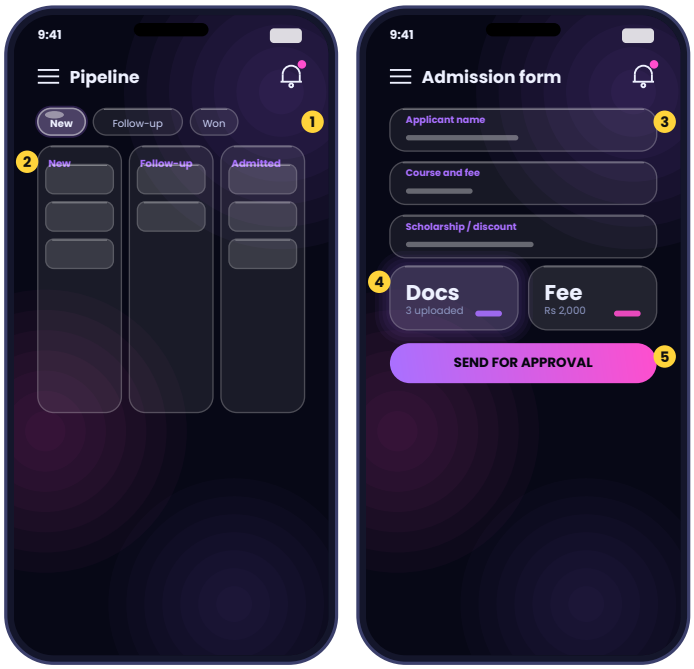
Figure 7.3 - Student Management: Students (left) and Add student (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F035	Students list (paged)	Screen	**[screen]** TAB 2: Students	Paged lazy list (Paging 3) with glass rows	GL	Stagger enter 40 ms, shimmer placeholders
F036	Search by name prefix	List	Students list (paged)	Glass search field, 300 ms debounce	GL	Field expand 250 ms, results fade-in
F037	Search by phone or enrolment number	List	Students list (paged)	Glass search field, 300 ms debounce	GL	Field expand 250 ms, results fade-in
F038	Filter by course	List	Students list (paged)	Filter chip row feeding a glass bottom sheet	LQ	Chip morph, sheet drag spring
F039	Filter by batch	List	Students list (paged)	Filter chip row feeding a glass bottom sheet	LQ	Chip morph, sheet drag spring
F040	Filter by status	List	Students list (paged)	Filter chip row feeding a glass bottom sheet	LQ	Chip morph, sheet drag spring
F041	Filter by fee-due state	List	Students list (paged)	Filter chip row feeding a glass bottom sheet	LQ	Chip morph, sheet drag spring
F042	Sort options	List	Students list (paged)	Sort menu in glass popup	GL	Popup scale-in from anchor 180 ms
F043	Add student form	Screen	**[screen]** TAB 2: Students	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F044	Duplicate phone/email check	Action	Add student form	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F045	Auto-generate enrolment number (transaction)	Action	Add student form	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

ID	Function	Type	Placed in	UI component	Mat	Animation
F046	Upload student photo	Action	Add student form	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F047	Upload ID proof	Action	Add student form	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F048	Student profile detail	Screen	**[screen]** TAB 2: Students	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F049	Edit student	Action	Student profile detail	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F050	Archive student	Action	Student profile detail	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F051	Restore archived student	Action	Student profile detail	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F052	Soft-delete student with audit	Action	Student profile detail	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F053	Assign student to course	Action	Student profile detail	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F054	Assign student to batch	Action	Student profile detail	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F055	Transfer student between batches	Action	Student profile detail	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F056	Link guardian	Action	Student profile detail	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F057	Student notes and remarks	Action	Student profile detail	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F058	Student document vault	Screen	**[screen]** TAB 2: Students	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F059	Generate student ID card	Action	Student document vault	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F060	Generate student QR code	Action	Student document vault	CameraX + ML Kit viewfinder with glass frame	GL	Scan-line sweep 1.6 s, success pulse + haptic
F061	Bulk import (CSV via Cloud Function)	Action	Student document vault	File picker + validation-result sheet	LQ	Progress bar, row-count ticker
F062	Bulk export (CSV/PDF)	Report	Student document vault	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F063	Bulk status change	Action	Student document vault	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F064	Student activity timeline	List	Student document vault	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms

M04 Admissions & Enquiries (20 functions)

Navigation: Drawer > People > Admissions **Roles:** Admin



Materials: Glass columns and cards, liquid source chips, clay convert button.

Motion: Drag lift/drop settle, stepper line fill, ring progress.

1 Pipeline board (F070) has three glass columns; cards lift (elevation + slight tilt) on drag and settle with a spring.

2 Source chips (F067) and counsellor assignment (F071) are liquid pills on each card.

3 Admission form (F074) is a stepper with glass fields and a progress line at top.

4 Document tiles (F075) show upload progress rings; fee capture (F076) uses tabular numerals.

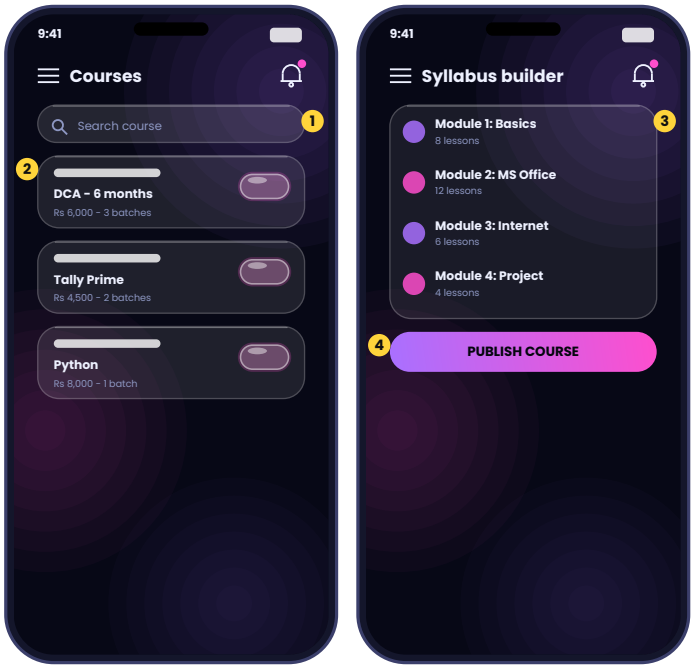
5 SEND FOR APPROVAL (F078) sits on a sticky glass bottom bar so it is always reachable.

Figure 7.4 - Admissions & Enquiries: Pipeline (left) and Admission form (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F065	Enquiry list	Screen	**[screen]** Menu > People > Admissions	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F066	Add enquiry	Action	Enquiry list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F067	Enquiry source tracking	Config	Enquiry list	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F068	Follow-up scheduler	Action	Enquiry list	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F069	Follow-up reminder notification	Job	Enquiry list	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F070	Enquiry pipeline board	Screen	**[screen]** Menu > People > Admissions	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring
F071	Assign enquiry to counsellor	Action	Enquiry pipeline board	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F072	Call log entry	Action	Enquiry pipeline board	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F073	Convert enquiry to admission	Action	Enquiry pipeline board	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F074	Admission form	Screen	**[screen]** Menu > People > Admissions	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F075	Upload admission documents	Action	Admission form	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F076	Admission fee capture at entry	Action	Admission form	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F077	Scholarship / discount apply	Action	Admission form	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F078	Admission approval workflow	Action	Admission form	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F079	Reject admission with reason	Action	Admission form	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F080	Waitlist management	Screen	**[screen]** Menu > People > Admissions	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F081	Admission confirmation message	Action	Waitlist management	Chat bubbles with composer	GL	Bubble pop-in spring, typing dots
F082	Enquiry-to-admission funnel report	Report	Waitlist management	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F083	Counsellor performance report	Report	Waitlist management	Report card with chart/table + share pill	GL	Draw-in 600 ms
F084	Duplicate enquiry detection	Job	Waitlist management	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift

M05 Course Catalog (18 functions)

Navigation: Drawer > Academics > Courses **Roles:** Admin (write); Teacher (read)



Materials: Glass cards, clay publish button, liquid status pills.

Motion: Reorder drag, card parallax, button press sink.

1 Course cards (F085) are glass with a thumbnail (F100) and a liquid status pill (published / draft, F101).

2 Add / edit / archive (F086-F088) live in the card overflow menu; archive uses hold-to-confirm.

3 Syllabus builder (F089, F090) is a reorderable list; drag handles appear on long-press and rows swap with a spring.

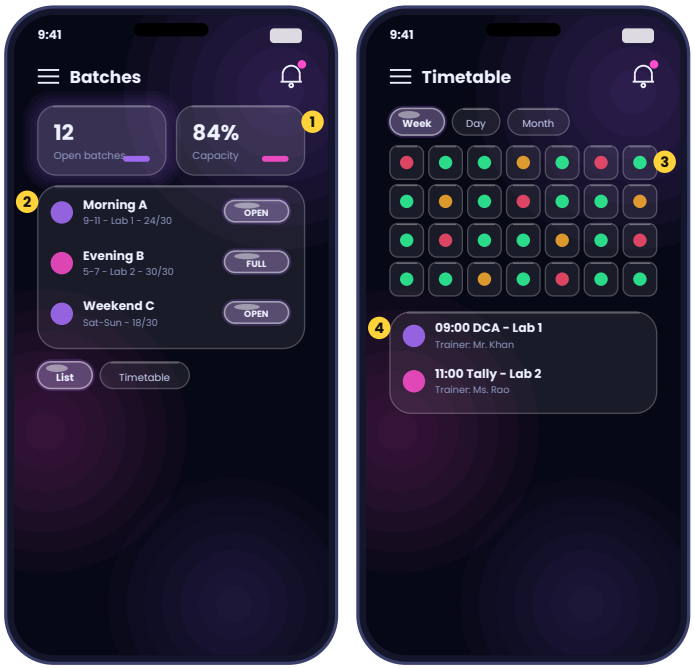
4 PUBLISH COURSE (F101) is a clay button that sinks 2 dp when pressed.

Figure 7.5 - Course Catalog: Courses (left) and Syllabus builder (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F085	Courses list	Screen	**[screen]** Menu > Academics > Courses	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F086	Add course	Action	Courses list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F087	Edit course	Action	Courses list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F088	Archive course	Action	Courses list	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F089	Course syllabus builder	Screen	**[screen]** Menu > Academics > Courses	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F090	Module and lesson ordering	Action	Course syllabus builder	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F091	Course duration and schedule rules	Config	Course syllabus builder	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F092	Course fee plan template	Config	Course syllabus builder	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F093	Course prerequisites	Config	Course syllabus builder	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F094	Course category management	Screen	**[screen]** Menu > Academics > Courses	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F095	Course eligibility rules	Config	Course category management	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F096	Certificate template link	Config	Course category management	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F097	Course demand heatmap	Report	Course category management	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F098	Course capacity limit	Config	Course category management	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F099	Course trainer mapping	Action	Course category management	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F100	Course thumbnail upload	Action	Course category management	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F101	Publish / unpublish course	Action	Course category management	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F102	Course version history	List	Course category management	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms

M06 Batch & Timetable (22 functions)

Navigation: Drawer > Academics > Batches **Roles:** Admin (write); Teacher (own batches); Student / Parent (read own)



Materials: Glass calendar cells, liquid view switcher, clay holiday badges.

Motion: Month slide, slot drag-resize, clash shake.

1 Capacity monitor (F107) is a glass tile; the ring colour moves from green to amber to red at 80 and 100 percent.

2 Batch rows (F103) show roster count and status; tap opens the roster (F108) with add / remove (F109, F110).

3 Calendar (F111) is a month grid; slots are small glass cards. Clash detection (F116) paints the slot border red and shakes it once.

4 Slot sheet (F112-F115) is a liquid bottom sheet with trainer and room pickers.

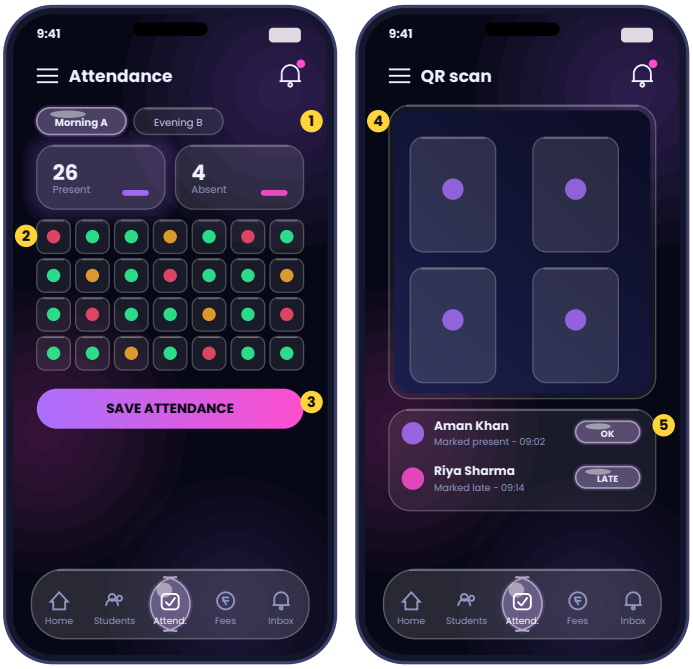
Figure 7.6 - Batch & Timetable: Batches (left) and Timetable (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F103	Batches list	Screen	**[screen]** Menu > Academics > Batches	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F104	Create batch	Action	Batches list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F105	Edit batch	Action	Batches list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F106	Close batch	Action	Batches list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F107	Batch capacity monitor	List	Batches list	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F108	Batch student roster	List	Batches list	Paged lazy list (Paging 3) with glass rows	GL	Stagger enter 40 ms, shimmer placeholders
F109	Add student to batch	Action	Batches list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F110	Remove student from batch	Action	Batches list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F111	Timetable calendar	Screen	**[screen]** Menu > Academics > Batches	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F112	Create timetable slot	Action	Timetable calendar	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F113	Recurring slot rule	Action	Timetable calendar	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F114	Room allocation	Action	Timetable calendar	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F115	Trainer allocation	Action	Timetable calendar	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F116	Clash detection (room/trainer)	Job	Timetable calendar	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F117	Reschedule class	Action	Timetable calendar	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F118	Cancel class with notice	Action	Timetable calendar	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F119	Holiday calendar	Screen	**[screen]** Menu > Academics > Batches	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F120	Add holiday	Action	Holiday calendar	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F121	Substitute trainer assign	Action	Holiday calendar	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F122	Timetable export (PDF)	Report	Holiday calendar	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up

ID	Function	Type	Placed in	UI component	Mat	Animation
F123	Timetable share to students	Action	Holiday calendar	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F124	Batch completion tracker	List	Holiday calendar	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms

M07 Attendance (20 functions)

Navigation: TAB 3: Attendance **Roles:** Admin; Teacher (own batches); Student / Parent (read own)



Materials: Glass tiles, clay roster cells, liquid batch chips.

Motion: Cell flip 180 ms, scan-line, success pulse, haptic tick per mark.

1 Batch chips (F133) switch roster; counters update live as cells change (F135).

2 Roster cells (F126-F130) are small clay-glass buttons: tap = present, long-press = late / excused; bulk bar (F130) marks all.

3 SAVE is a neon CTA; offline marks go to the WorkManager queue (F144) and show a "queued" chip.

4 QR scanner (F127) uses CameraX + ML Kit; a glass frame, scan-line sweep and a green pulse on success.

5 Recent scans list slides in under the viewfinder; duplicate scans show a soft shake.

Figure 7.7 - Attendance: Attendance (left) and QR scan (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F125	Attendance home (today)	Screen	**[screen]** TAB 3: Attendance	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F126	Mark student attendance	Action	Attendance home (today)	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F127	Mark with QR scan	Action	Attendance home (today)	CameraX + ML Kit viewfinder with glass frame	GL	Scan-line sweep 1.6 s, success pulse + haptic
F128	Mark with roster tap	Action	Attendance home (today)	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F129	Mark late / excused	Action	Attendance home (today)	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F130	Bulk mark all present	Action	Attendance home (today)	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F131	Edit attendance (reason + audit)	Action	Attendance home (today)	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F132	Lock attendance after cut-off	Job	Attendance home (today)	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F133	Attendance by batch	List	Attendance home (today)	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F134	Attendance by student	List	Attendance home (today)	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F135	Attendance percentage calculation	Job	Attendance home (today)	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F136	Low-attendance alert	Job	Attendance home (today)	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F137	Parent absence notice	Job	Attendance home (today)	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F138	Staff attendance mark	Action	Attendance home (today)	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F139	Staff attendance by QR / geo-fence	Action	Attendance home (today)	CameraX + ML Kit viewfinder with glass frame	GL	Scan-line sweep 1.6 s, success pulse + haptic
F140	Leave-linked attendance sync	Job	Attendance home (today)	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift

ID	Function	Type	Placed in	UI component	Mat	Animation
F141	Attendance monthly report	Report	Attendance home (today)	Report card with chart/table + share pill	GL	Draw-in 600 ms
F142	Attendance heatmap	Report	Attendance home (today)	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F143	Attendance export	Report	Attendance home (today)	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F144	Offline attendance queue (WorkManager)	Job	Attendance home (today)	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift

M08 Fees & Payments (28 functions)

Navigation: TAB 4: Fees **Roles:** Admin (write); Parent / Student (read own)



Materials: Glass tiles/rows, clay primary CTA, liquid mode chips.

Motion: Ticker roll, tick Lottie, share-sheet slide.

1 Collection tiles (F165) use tabular numerals; the amount rolls digit by digit when it changes.

2 Mode chart (F166) and course chart (F167) are glass with draw-in animation.

3 Recent payments (F155, F156) are glass rows with right-aligned amounts and a mode chip.

4 Payment form (F149-F151): the student field is a search sheet; amount uses a numeric pad with haptics.

5 Receipt preview (F153) is a document card with share pill (F154).

6 POST PAYMENT posts through the server function; the button shows progress then a Lottie tick.

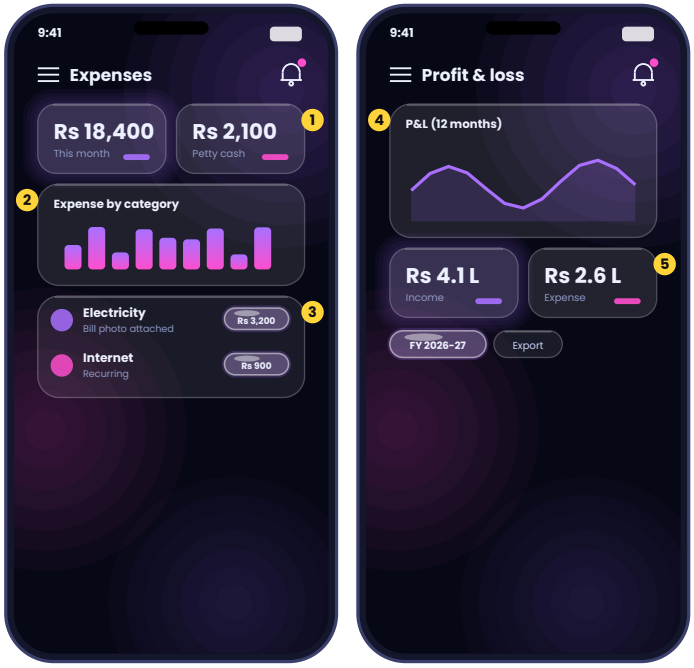
Figure 7.8 - Fees & Payments: Fees (left) and Record payment (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F145	Fees home	Screen	**[screen]** TAB 4: Fees	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F146	Fee structure per course	Config	Fees home	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F147	Instalment plan builder	Action	Fees home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F148	Assign fee plan to student	Action	Fees home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F149	Record cash payment	Action	Fees home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F150	Record bank / UPI payment	Action	Fees home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F151	Online payment via gateway (server verified)	Action	Fees home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F152	Payment webhook verification	Job	Fees home	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F153	Generate receipt	Action	Fees home	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F154	Share receipt PDF	Action	Fees home	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F155	Fee due list	List	Fees home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F156	Overdue list	List	Fees home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F157	Scheduled due reminders	Job	Fees home	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F158	Apply discount	Action	Fees home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

ID	Function	Type	Placed in	UI component	Mat	Animation
F159	Late-fee rule	Config	Fees home	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F160	Refund request	Action	Fees home	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F161	Approve refund	Action	Fees home	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F162	Fee waiver	Action	Fees home	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F163	Concession approval workflow	Action	Fees home	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F164	Fee ledger per student	Screen	**[screen]** TAB 4: Fees	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F165	Daily collection summary	Report	Fee ledger per student	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F166	Collection by payment mode	Report	Fee ledger per student	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F167	Collection by course	Report	Fee ledger per student	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F168	Collection by staff	Report	Fee ledger per student	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F169	Fee defaulters report	Report	Fee ledger per student	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F170	Cheque / bounce tracking	Action	Fee ledger per student	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F171	Fee audit trail	List	Fee ledger per student	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F172	Fee data export	Report	Fee ledger per student	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms

M09 Accounts & Expenses (18 functions)

Navigation: Drawer > Finance > Accounts **Roles:** Admin



Materials: Glass charts and rows, liquid export pill, clay approve button.

Motion: Draw-in, tooltip spring, pill-to-progress morph.

1 Expense tiles and chart (F173, F175) are glass; categories colour-code with the semantic palette.

2 Add expense (F174) opens a liquid sheet; bill photo (F176) uses CameraX with auto-crop.

3 Vendor and petty-cash screens (F177-F182) share the same row pattern for consistency.

4 P&L; (F184) is a line chart with area fill; hovering a month shows a glass tooltip.

5 Export pill (F188 and others) morphs to a progress bar then opens the share sheet.

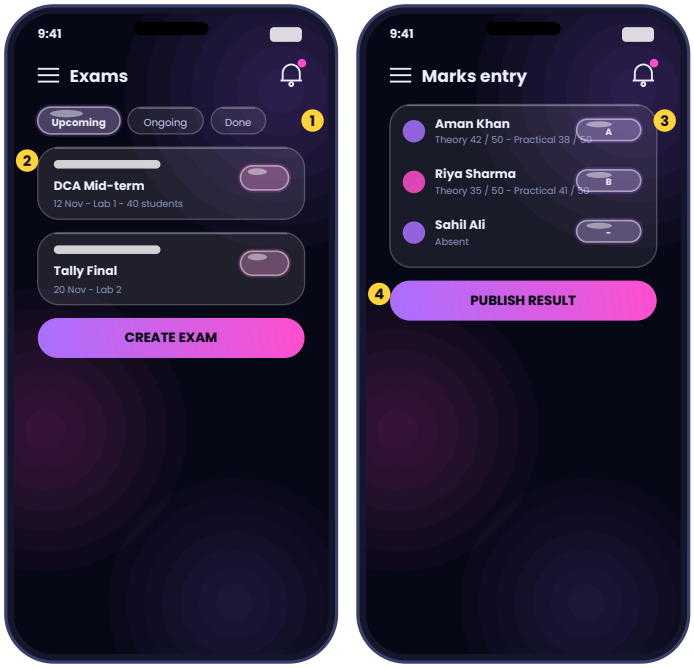
Figure 7.9 - Accounts & Expenses: Expenses (left) and Profit & loss (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F173	Expense list	Screen	**[screen]** Menu > Finance > Accounts	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F174	Add expense	Action	Expense list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F175	Expense categories	Config	Expense list	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F176	Attach bill photo	Action	Expense list	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F177	Vendor directory	Screen	**[screen]** Menu > Finance > Accounts	Paged lazy list (Paging 3) with glass rows	GL	Stagger enter 40 ms, shimmer placeholders

ID	Function	Type	Placed in	UI component	Mat	Animation
F178	Add vendor	Action	Vendor directory	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F179	Vendor payment	Action	Vendor directory	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F180	Recurring expense schedule	Action	Vendor directory	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F181	Petty cash register	Screen	**[screen]** Menu > Finance > Accounts	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F182	Petty cash top-up	Action	Petty cash register	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F183	Non-fee income entry	Action	Petty cash register	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F184	Profit and loss report	Report	Petty cash register	Report card with chart/table + share pill	GL	Draw-in 600 ms
F185	Monthly balance view	Report	Petty cash register	Report card with chart/table + share pill	GL	Draw-in 600 ms
F186	Budget plan	Config	Petty cash register	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F187	Budget vs actual	Report	Petty cash register	Report card with chart/table + share pill	GL	Draw-in 600 ms
F188	Tax / GST summary report	Report	Petty cash register	Report card with chart/table + share pill	GL	Draw-in 600 ms
F189	Expense approval	Action	Petty cash register	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F190	Financial year close	Action	Petty cash register	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

M10 Examinations & Results (24 functions)

Navigation: Drawer > Academics > Exams **Roles:** Admin; Teacher (own batches); Student / Parent (read own)



Materials: Glass cards and rows, clay publish button, liquid grade chips.

Motion: Countdown ring, grade chip recolour, confetti on publish.

1 Exam cards (F191-F193) show schedule and invigilator chips (F213); the countdown ring pulses in the last hour.

2 Question bank (F194-F197) uses search plus liquid category chips; CSV import (F196) shows a validation sheet.

3 Marks grid (F202) is a spreadsheet-style glass list; the grade chip recolours as marks change (F204).

4 PUBLISH RESULT (F206) demands confirmation; a success burst plays and students receive a push.

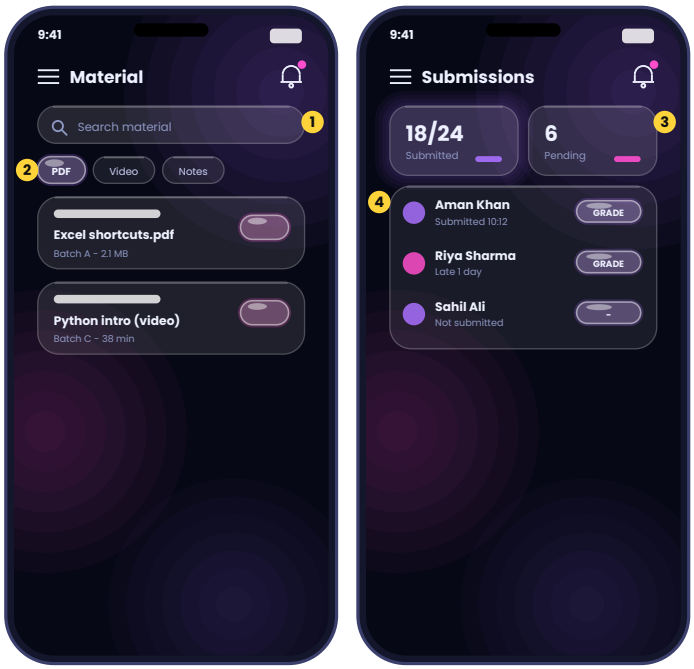
Figure 7.10 - Examinations & Results: Exams (left) and Marks entry (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F191	Exams list	Screen	**[screen]** Menu > Academics > Exams	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F192	Create exam	Action	Exams list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F193	Exam schedule	Action	Exams list	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F194	Question bank	Screen	**[screen]** Menu > Academics > Exams	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F195	Add question	Action	Question bank	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F196	Import questions (CSV)	Action	Question bank	File picker + validation-result sheet	LQ	Progress bar, row-count ticker

ID	Function	Type	Placed in	UI component	Mat	Animation
F197	Paper builder	Action	Question bank	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F198	Online test attempt (student)	Screen	**[screen]** Menu > Academics > Exams	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F199	Timer and autosave of attempt	Job	Online test attempt (student)	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F200	Auto-evaluate objective answers (server)	Job	Online test attempt (student)	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F201	Manual marking screen	Screen	**[screen]** Menu > Academics > Exams	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F202	Marks entry grid	Action	Manual marking screen	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F203	Grace-marks rule	Config	Manual marking screen	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F204	Grade scale config	Config	Manual marking screen	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F205	Result calculation	Job	Manual marking screen	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F206	Result publish / hold	Action	Manual marking screen	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F207	Re-evaluation request	Action	Manual marking screen	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F208	Rank list	Report	Manual marking screen	Report card with chart/table + share pill	GL	Draw-in 600 ms
F209	Subject-wise analysis	Report	Manual marking screen	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F210	Student result card PDF	Report	Manual marking screen	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F211	Bulk result export	Report	Manual marking screen	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F212	Hall ticket generate	Action	Manual marking screen	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F213	Invigilator assignment	Action	Manual marking screen	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F214	Exam attendance	Action	Manual marking screen	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

M11 Assignments & Study Material (18 functions)

Navigation: Drawer > Academics > Material **Roles:** Admin; Teacher (write); Student (read assigned)



Materials: Glass cards, liquid type chips, clay grade button.

Motion: Upload ring, card enter, sheet drag.

1 Library cards (F215-F219) show type icon, size and visibility; video items stream via Media3 with cache.

2 Upload (F216) offers PDF, video, notes; progress ring and WorkManager keep uploads alive in background.

3 Status tiles (F229) summarise submissions; late items carry an amber chip (F223).

4 Grade action (F225) opens a glass sheet with feedback comments (F226).

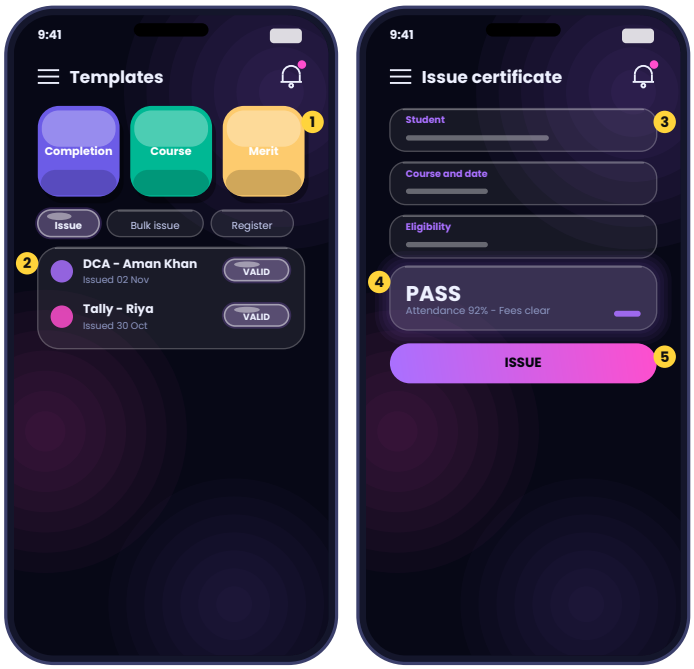
Figure 7.11 - Assignments & Study Material: Material (left) and Submissions (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F215	Material library	Screen	**[screen]** Menu > Academics > Material	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms

ID	Function	Type	Placed in	UI component	Mat	Animation
F216	Upload PDF / video / notes	Action	Material library	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F217	Material categories	Config	Material library	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F218	Assign material to batch	Action	Material library	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F219	Visibility and expiry rules	Config	Material library	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F220	Download tracking	List	Material library	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F221	Assignments list	Screen	**[screen]** Menu > Academics > Material	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F222	Create assignment	Action	Assignments list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F223	Due-date and late policy	Config	Assignments list	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F224	Student submission upload	Action	Assignments list	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F225	Grade submission	Action	Assignments list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F226	Feedback comments	Action	Assignments list	Chat bubbles with composer	GL	Bubble pop-in spring, typing dots
F227	Similarity flag on submissions (server)	Job	Assignments list	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F228	Allow resubmission	Action	Assignments list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F229	Submission status board	List	Assignments list	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring
F230	Attach recorded lecture	Action	Assignments list	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F231	Material search	List	Assignments list	Glass search field, 300 ms debounce	GL	Field expand 250 ms, results fade-in
F232	Storage usage per batch	Report	Assignments list	Report card with chart/table + share pill	GL	Draw-in 600 ms

M12 Certificates (14 functions)

Navigation: Drawer > Academics > Certificates **Roles:** Admin; Student (read own)



Materials: Clay template tiles, glass forms, neon eligibility edge.

Motion: Tile lift, PDF fade-through, stamp-in animation on issue.

1 Template tiles (F234) use clay for a tactile, certificate-like feel; selection lifts the tile.

2 Register (F246) lists issued certificates with a QR (F239) and a VALID / REVOKED chip.

3 Issue form (F236) shows eligibility (F245) automatically: attendance, results, fees.

4 Eligibility result is a glass tile with a green or red neon edge.

5 ISSUE renders the PDF (F243) on the server, then previews it with a share pill (F244).

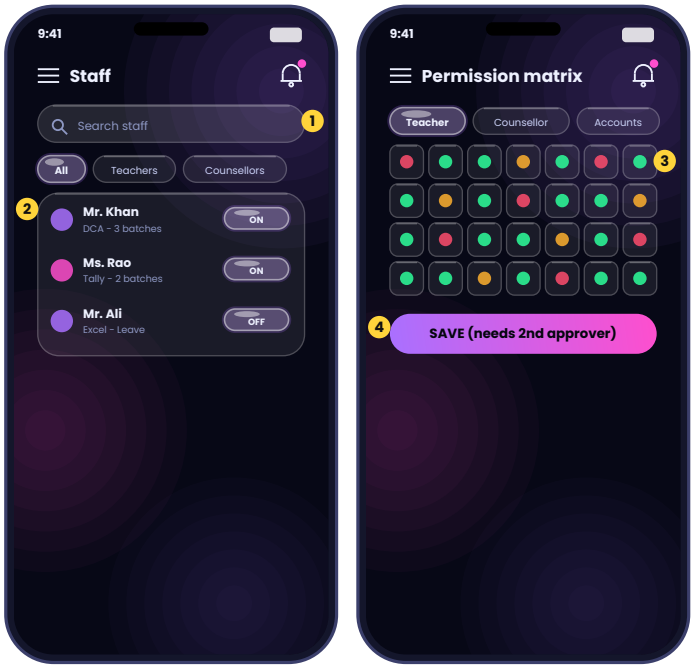
Figure 7.12 - Certificates: Templates (left) and Issue certificate (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F233	Certificates home	Screen	**[screen]** Menu > Academics > Certificates	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms

ID	Function	Type	Placed in	UI component	Mat	Animation
F234	Certificate templates	Screen	**[screen]** Menu > Academics > Certificates	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F235	Template field editor	Action	Certificate templates	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F236	Issue certificate	Action	Certificate templates	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F237	Bulk issue to batch	Action	Certificate templates	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F238	Certificate number sequence	Job	Certificate templates	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F239	QR verification code	Action	Certificate templates	CameraX + ML Kit viewfinder with glass frame	GL	Scan-line sweep 1.6 s, success pulse + haptic
F240	Public verify endpoint (callable)	Job	Certificate templates	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F241	Revoke certificate	Action	Certificate templates	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F242	Reissue certificate	Action	Certificate templates	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F243	Certificate PDF render	Action	Certificate templates	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F244	Share certificate	Action	Certificate templates	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F245	Eligibility check (attendance + results + fees)	Job	Certificate templates	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F246	Issued certificates register	Report	Certificate templates	Report card with chart/table + share pill	GL	Draw-in 600 ms

M13 Teacher & Staff Management (22 functions)

Navigation: Drawer > People > Staff **Roles:** Admin



Materials: Glass rows, glass switches, clay approve.

Motion: Switch spring, row expand, approval stamp.

1 Staff list (F247) shows availability status; invite (F265) and reset (F266) live in the row menu.

2 Workload (F255) is a small bar inside each row.

3 Permission matrix (F253) is a grid of glass switches per role and capability.

4 Role change (F252) is four-eyes: a second admin approves, and an audit row is written.

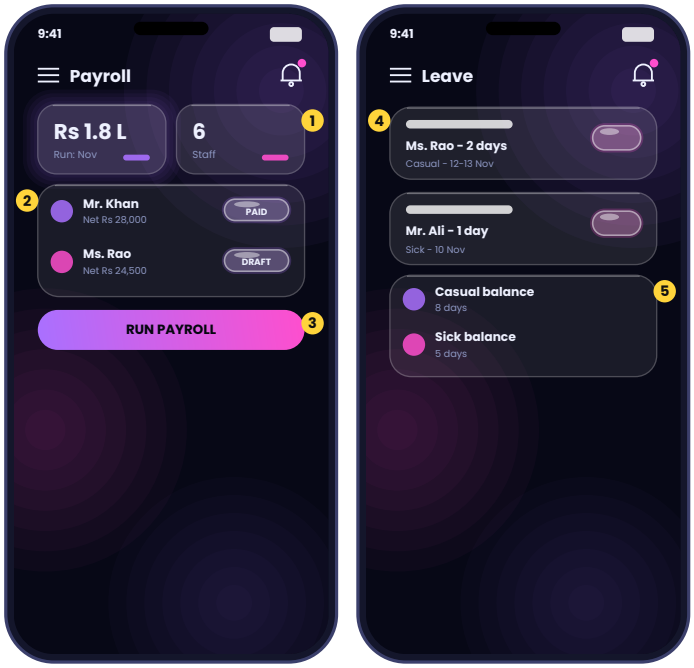
Figure 7.13 - Teacher & Staff Management: Staff (left) and Permission matrix (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F247	Staff list	Screen	**[screen]** Menu > People > Staff	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F248	Add staff	Action	Staff list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F249	Edit staff	Action	Staff list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F250	Deactivate staff	Action	Staff list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F251	Staff profile	Screen	**[screen]** Menu > People > Staff	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms

ID	Function	Type	Placed in	UI component	Mat	Animation
F252	Role assignment (custom claim)	Action	Staff profile	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F253	Permission matrix editor	Screen	**[screen]** Menu > People > Staff	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F254	Assign batches to teacher	Action	Permission matrix editor	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F255	Teacher workload view	List	Permission matrix editor	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F256	Staff documents	Action	Permission matrix editor	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F257	Staff photo	Action	Permission matrix editor	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F258	Qualification records	Action	Permission matrix editor	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F259	Onboarding checklist	Action	Permission matrix editor	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring
F260	Offboarding checklist	Action	Permission matrix editor	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring
F261	Staff directory search	List	Permission matrix editor	Glass search field, 300 ms debounce	GL	Field expand 250 ms, results fade-in
F262	Staff performance review	Action	Permission matrix editor	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F263	Student feedback on teacher view	List	Permission matrix editor	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F264	Teacher availability calendar	Screen	**[screen]** Menu > People > Staff	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F265	Invite staff (email link)	Action	Teacher availability calendar	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F266	Admin-initiated staff password reset	Action	Teacher availability calendar	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms
F267	Staff import	Action	Teacher availability calendar	File picker + validation-result sheet	LQ	Progress bar, row-count ticker
F268	Staff export	Report	Teacher availability calendar	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms

M14 Payroll & Leave (16 functions)

Navigation: Drawer > Finance > Payroll **Roles:** Admin; Staff (own payslip, leave)



Materials: Glass tiles, clay run button, liquid share pill.

Motion: Lock badge drop, ticker, approval stamp.

1 Payroll tiles (F269-F272) use tabular numerals; locking a month (F284) greys the screen with a lock badge.

2 Payslip rows (F273, F274) open a document preview; share is a liquid pill.

3 RUN PAYROLL requires re-authentication and four-eyes approval.

4 Leave requests (F277, F278) are approval cards with Approve / Reject.

5 Balances (F279) are glass rows with a thin progress bar.

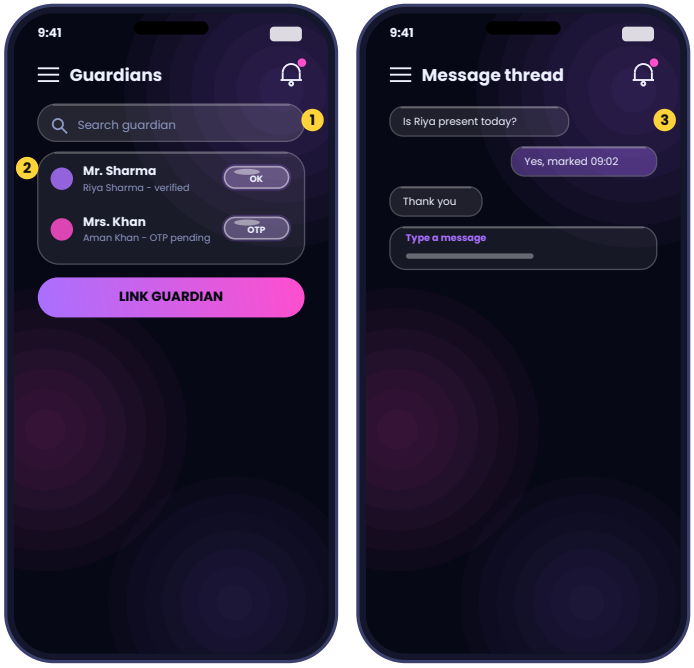
Figure 7.14 - Payroll & Leave: Payroll (left) and Leave (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F269	Payroll home	Screen	**[screen]** Menu > Finance > Payroll	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F270	Salary structure	Config	Payroll home	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s

ID	Function	Type	Placed in	UI component	Mat	Animation
F271	Assign salary to staff	Action	Payroll home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F272	Monthly payroll run	Action	Payroll home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F273	Generate payslip	Action	Payroll home	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F274	Share payslip	Action	Payroll home	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F275	Deductions and advances	Action	Payroll home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F276	Leave types	Config	Payroll home	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F277	Leave request (staff)	Action	Payroll home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F278	Leave approval	Action	Payroll home	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F279	Leave balance view	List	Payroll home	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F280	Holiday-linked leave rule	Config	Payroll home	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F281	Payroll register report	Report	Payroll home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F282	Payroll bank-file export	Report	Payroll home	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F283	Overtime entry	Action	Payroll home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F284	Lock payroll month	Action	Payroll home	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms

M15 Guardian (Parent) Linking (12 functions)

Navigation: Drawer > People > Guardians **Roles:** Admin; Parent (own profile)



Materials: Glass bubbles, liquid dialogs, clay link button.

Motion: Bubble pop-in, OTP box focus, confirm dialog morph.

1 Guardian list (F285) shows verification state; OTP (F287) uses a 6-box glass input with auto-advance.

2 Link / unlink (F288, F289) are confirm dialogs in liquid style.

3 Thread (F294) uses glass bubbles; own messages tint with the accent.

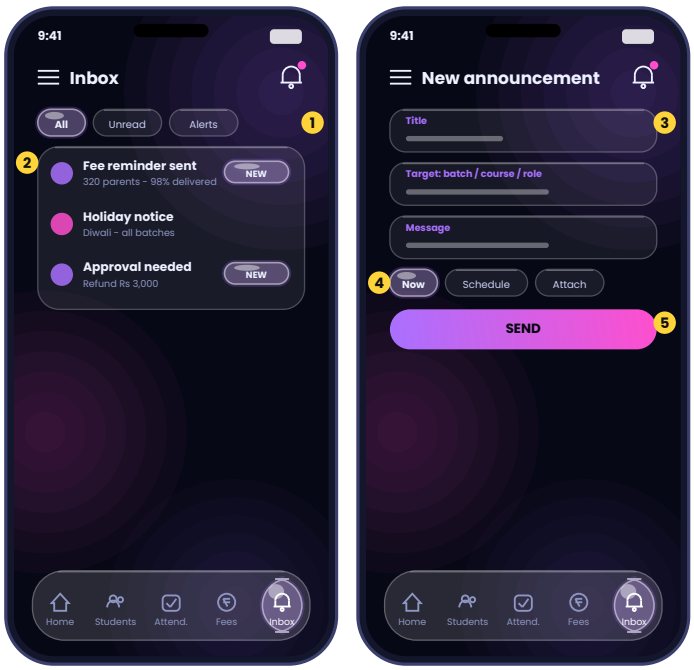
Figure 7.15 - Guardian (Parent) Linking: Guardians (left) and Message thread (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F285	Guardian list	Screen	**[screen]** Menu > People > Guardians	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F286	Add guardian	Action	Guardian list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F287	Verify guardian phone (OTP)	Action	Guardian list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F288	Link guardian to student	Action	Guardian list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F289	Unlink guardian	Action	Guardian list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

ID	Function	Type	Placed in	UI component	Mat	Animation
F290	Guardian visibility rules	Config	Guardian list	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F291	Parent notification preferences	Config	Guardian list	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F292	Parent fee-view access	Config	Guardian list	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F293	Parent result-view access	Config	Guardian list	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F294	Guardian message thread	Screen	**[screen]** Menu > People > Guardians	Chat bubbles with composer	GL	Bubble pop-in spring, typing dots
F295	Guardian consent record	Action	Guardian message thread	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F296	Guardian invite link	Action	Guardian message thread	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

M16 Notifications & Announcements (18 functions)

Navigation: TAB 5: Inbox **Roles:** Admin (send); Teacher (send to own batches); all receive



Materials: Glass rows, liquid pills, clay emergency button.

Motion: Swipe actions, banner slide-down, hold-to-confirm ring.

1 Inbox (F303) with unread dot; swipe right marks read (F304), swipe left archives.

2 Delivery tracking (F305) is a thin segmented bar inside the row.

3 Targeting (F299) uses chips for batch, course and role with audience count preview.

4 Schedule and attach (F300, F313) are liquid pills.

5 SEND triggers the FCM function; emergency broadcast (F310) uses a red clay button with hold-to-confirm.

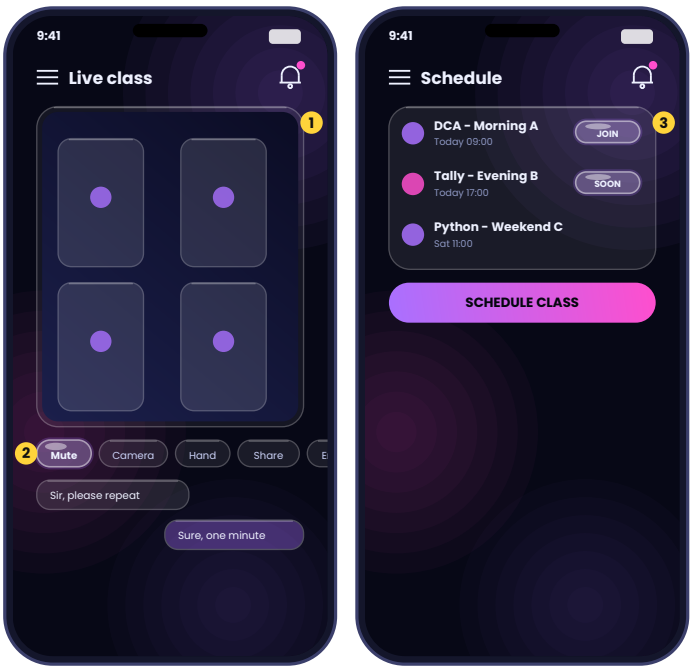
Figure 7.16 - Notifications & Announcements: Inbox (left) and New announcement (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F297	Announcements list	Screen	**[screen]** TAB 5: Inbox	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F298	Create announcement	Action	Announcements list	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F299	Target by batch / course / role	Action	Announcements list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F300	Schedule announcement	Action	Announcements list	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F301	Push notification send (FCM via function)	Action	Announcements list	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F302	Notification templates	Config	Announcements list	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F303	In-app inbox	Screen	**[screen]** TAB 5: Inbox	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F304	Mark read / unread	Action	In-app inbox	Roster cell (Present / Absent / Late) + bulk bar	CL	Cell flip 180 ms, haptic tick
F305	Delivery status tracking	List	In-app inbox	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F306	Topic subscription management	Job	In-app inbox	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F307	Birthday greeting automation	Job	In-app inbox	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift

ID	Function	Type	Placed in	UI component	Mat	Animation
F308	Fee reminder template	Config	In-app inbox	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F309	SMS / WhatsApp hand-off link	Action	In-app inbox	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F310	Emergency broadcast	Action	In-app inbox	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F311	Quiet-hours rules	Config	In-app inbox	Settings switch / slider row	GL	Thumb spring, haptic confirm
F312	Notification history	List	In-app inbox	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F313	Attachment to announcement	Action	In-app inbox	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F314	Announcement reach analytics	Report	In-app inbox	Banner / inbox row / push composer	GL	Banner slide-down 250 ms

M17 Live Classes (Agora) (20 functions)

Navigation: Drawer > Academics > Live Classes **Roles:** Admin; Teacher (host); Student (join assigned)



Materials: Liquid glass control bar over video, glass chat, clay End button.

Motion: Controls auto-hide, mic ripple, join transition (shared element).

1 Video grid (F317, F318) renders Agora views; control bar (F321-F326) is liquid glass and auto-hides after 3 s.

2 Mic button shows a ripple with audio level; hand-raise (F323) bounces once.

3 Schedule (F316) shows join state; the token is minted just in time (F319, F320).

Figure 7.17 - Live Classes (Agora): Live class (left) and Schedule (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F315	Live classes home	Screen	**[screen]** Menu > Academics > Live Classes	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F316	Schedule live class	Action	Live classes home	Calendar / week grid with slot cards	GL	Day-cell ripple, month slide 300 ms
F317	Start live class (teacher)	Action	Live classes home	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F318	Join live class (student)	Action	Live classes home	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F319	Server-minted Agora token	Job	Live classes home	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F320	Channel access check (role + batch)	Job	Live classes home	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F321	Camera / microphone permission flow	Action	Live classes home	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F322	Mute / unmute participant	Action	Live classes home	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F323	Raise hand	Action	Live classes home	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F324	Screen share	Action	Live classes home	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F325	In-class chat	Action	Live classes home	Chat bubbles with composer	GL	Bubble pop-in spring, typing dots

ID	Function	Type	Placed in	UI component	Mat	Animation
F326	Recording trigger	Action	Live classes home	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F327	Recordings list	Screen	**[screen]** Menu > Academics > Live Classes	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F328	Auto-attendance from join log	Job	Recordings list	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F329	Remove participant	Action	Recordings list	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F330	Network quality indicator	List	Recordings list	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F331	Low-bandwidth audio-only mode	Action	Recordings list	Agora call controls bar	LQ	Controls auto-hide 3 s, mic ripple
F332	Class end summary	Report	Recordings list	Report card with chart/table + share pill	GL	Draw-in 600 ms
F333	Live class reminders	Job	Recordings list	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F334	Concurrent channel usage monitor	Report	Recordings list	Report card with chart/table + share pill	GL	Draw-in 600 ms

M18 AI Tutor & Assistant (18 functions)

Navigation: Drawer > Academics > AI Hub **Roles:** Student / Teacher (tutor); Admin (assistant, config)



Materials: Glass chat, glass tiles, liquid provider chips, clay kill switch.

Motion: Streaming reveal, typing dots, switch spring.

1 Chat (F336) streams text word by word; the composer supports image / PDF attachment (F337).

2 Quota bar (F341) sits above the composer; at limit the composer disables with a clear message.

3 Usage dashboard (F350) shows requests and cost per provider.

4 Kill switch (F352) is a red clay switch; flipping it shows a confirm and takes effect instantly.

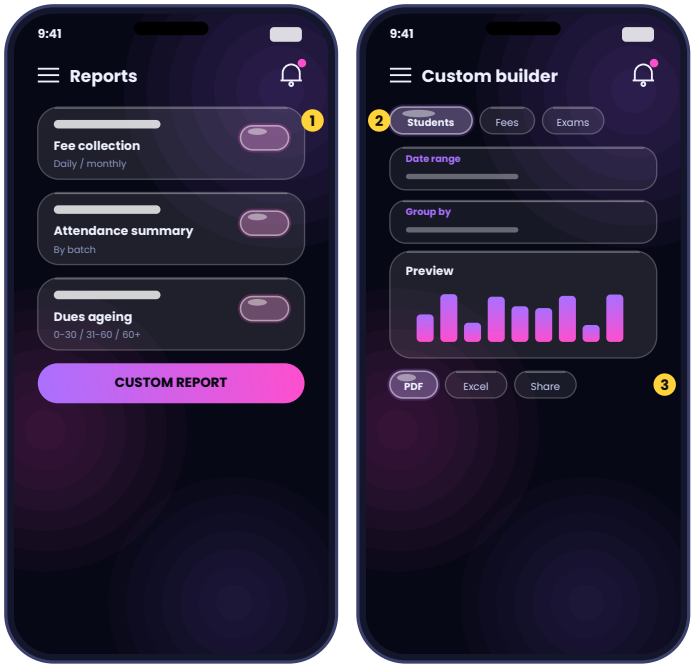
Figure 7.18 - AI Tutor & Assistant: AI tutor (left) and AI usage (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F335	AI hub	Screen	**[screen]** Menu > Academics > AI Hub	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F336	Student AI tutor chat	Screen	**[screen]** Menu > Academics > AI Hub	Chat bubbles with composer	GL	Bubble pop-in spring, typing dots
F337	Ask with attachment (image / PDF)	Action	Student AI tutor chat	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F338	Server proxy call (Gemini)	Job	Student AI tutor chat	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F339	Server proxy call (OpenAI)	Job	Student AI tutor chat	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F340	Provider routing and fallback	Job	Student AI tutor chat	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F341	Per-user quota and rate limits	Job	Student AI tutor chat	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F342	Prompt templates (admin)	Config	Student AI tutor chat	AI chat / admin AI panel	GL	Streaming text reveal, typing dots

ID	Function	Type	Placed in	UI component	Mat	Animation
F343	Safety and content filter	Job	Student AI tutor chat	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F344	Conversation history	List	Student AI tutor chat	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F345	Delete conversation	Action	Student AI tutor chat	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F346	Admin AI assistant (questions over reports)	Screen	**[screen]** Menu > Academics > AI Hub	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F347	AI-generated quiz from material	Action	Admin AI assistant (questions over	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F348	AI-generated lesson summary	Action	Admin AI assistant (questions over	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F349	AI answer feedback	Action	Admin AI assistant (questions over	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F350	Usage and cost dashboard	Report	Admin AI assistant (questions over	Kanban board with draggable cards	GL	Drag lift (elevation + 3 deg tilt), drop settle spring
F351	Provider key-rotation trigger	Config	Admin AI assistant (questions over	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F352	AI kill switch	Config	Admin AI assistant (questions over	AI chat / admin AI panel	GL	Streaming text reveal, typing dots

M19 Reports & Analytics (22 functions)

Navigation: Drawer > Insights > Reports **Roles:** Admin; Teacher (own batches, limited)



Materials: Glass cards and charts, liquid entity/export chips.

Motion: Sparkline draw-in, preview cross-fade, drill-down zoom.

1 Report library (F353-F363) is a list of glass cards with a thumbnail sparkline.

2 Builder (F364-F371): choose entity chips, range, grouping; preview chart updates live.

3 Export pills (F367-F369) generate PDF or Excel through functions and then open the share sheet.

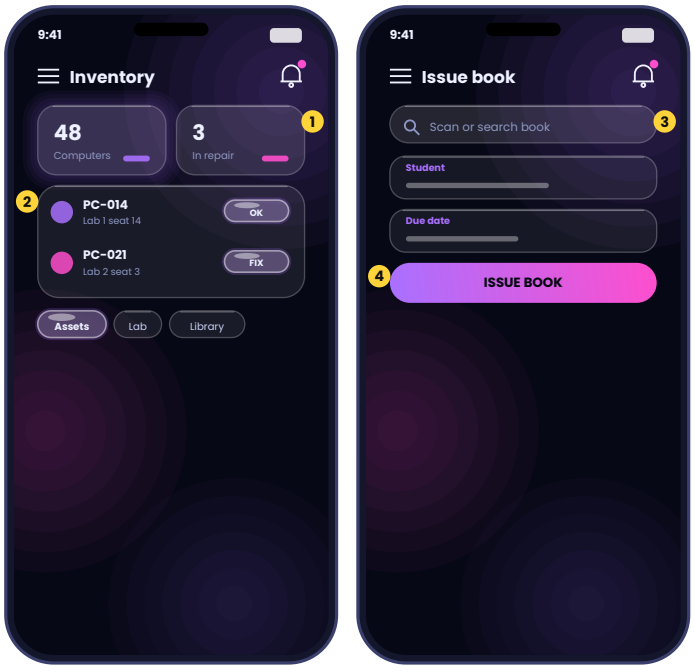
Figure 7.19 - Reports & Analytics: Reports (left) and Custom builder (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F353	Reports home	Screen	**[screen]** Menu > Insights > Reports	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F354	Student strength report	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F355	Admission report	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F356	Course-wise enrolment	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F357	Batch-wise report	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F358	Attendance summary report	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F359	Fee collection report	Report	Reports home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change
F360	Dues ageing report	Report	Reports home	Money row/card with tabular numerals	GL	Ticker roll 500 ms on amount change

ID	Function	Type	Placed in	UI component	Mat	Animation
F361	Exam performance report	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F362	Teacher workload report	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F363	Dropout-risk list	Report	Reports home	Report card with chart/table + share pill	GL	Draw-in 600 ms
F364	Custom report builder	Screen	**[screen]** Menu > Insights > Reports	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F365	Save report preset	Action	Custom report builder	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F366	Schedule report email	Job	Custom report builder	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F367	Export to PDF	Report	Custom report builder	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F368	Export to Excel / CSV	Report	Custom report builder	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F369	Share report	Action	Custom report builder	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F370	Chart drill-down	Action	Custom report builder	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F371	Date-range comparison	Action	Custom report builder	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F372	Branch comparison (multi-branch)	Report	Custom report builder	Vico chart inside GlassCard	GL	Series draw-in 700 ms, tooltip spring
F373	Data freshness indicator	List	Custom report builder	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F374	BigQuery export toggle	Config	Custom report builder	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms

M20 Inventory, Lab & Library (14 functions)

Navigation: Drawer > Operations > Inventory **Roles:** Admin



Materials: Glass tiles and rows, clay issue button.

Motion: Scan sweep, chip pulse, timeline draw.

1 Asset tiles (F375-F377) with QR labels (F387); low-stock chip pulses amber (F386).

2 Lab register (F378-F380) ties asset to seat; maintenance log is a timeline.

3 Library issue (F382) scans a QR; fines (F384) are calculated by a job.

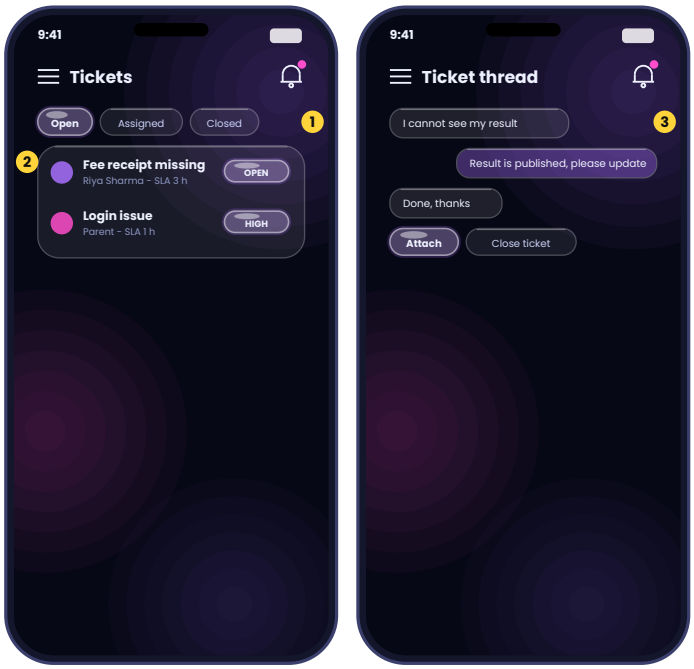
Figure 7.20 - Inventory, Lab & Library: Inventory (left) and Issue book (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F375	Inventory home	Screen	**[screen]** Menu > Operations > Inventory	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F376	Add asset	Action	Inventory home	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F377	Asset categories	Config	Inventory home	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s

ID	Function	Type	Placed in	UI component	Mat	Animation
F378	Lab computer register	Screen	**[screen]** Menu > Operations > Inventory	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F379	Assign asset to lab seat	Action	Lab computer register	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F380	Maintenance log	Action	Lab computer register	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F381	Library books list	Screen	**[screen]** Menu > Operations > Inventory	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F382	Issue book	Action	Library books list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F383	Return book	Action	Library books list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F384	Fine calculation	Job	Library books list	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift
F385	Stock count	Action	Library books list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F386	Low-stock alert	Job	Library books list	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F387	Asset QR labels	Action	Library books list	CameraX + ML Kit viewfinder with glass frame	GL	Scan-line sweep 1.6 s, success pulse + haptic
F388	Asset report	Report	Library books list	Report card with chart/table + share pill	GL	Draw-in 600 ms

M21 Support & Feedback (12 functions)

Navigation: Drawer > Operations > Support **Roles:** Admin; Teacher; Student / Parent (create, read own)



Materials: Glass rows, clay stars, liquid status chips.

Motion: SLA ring colour shift, bubble pop-in.

1 Tickets (F389-F392) show SLA timer (F400); the timer ring turns amber then red.

2 Thread (F393) supports screenshot attach (F394).

3 Feedback surveys (F395-F397) use clay star rating buttons.

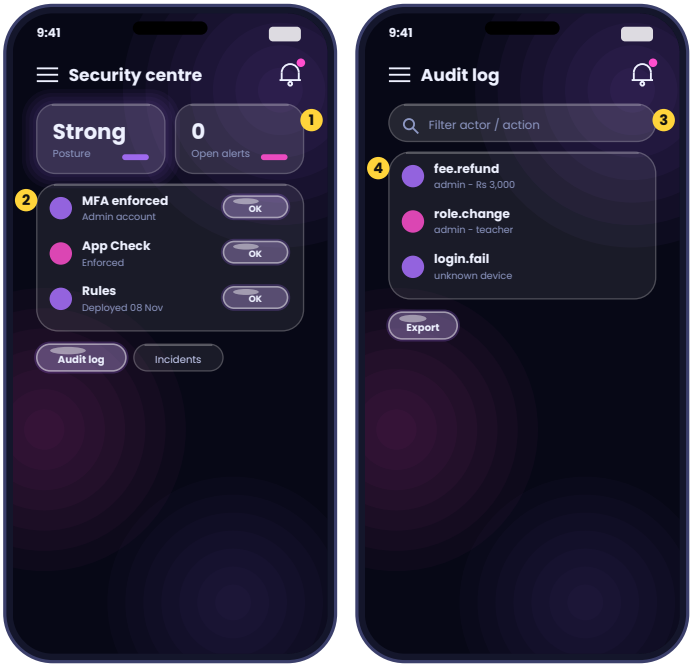
Figure 7.21 - Support & Feedback: Tickets (left) and Ticket thread (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F389	Support tickets list	Screen	**[screen]** Menu > Operations > Support	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F390	Create ticket	Action	Support tickets list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F391	Assign ticket	Action	Support tickets list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F392	Update ticket status	Action	Support tickets list	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F393	Ticket chat thread	Screen	**[screen]** Menu > Operations > Support	Chat bubbles with composer	GL	Bubble pop-in spring, typing dots
F394	Attach screenshot	Action	Ticket chat thread	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms

ID	Function	Type	Placed in	UI component	Mat	Animation
F395	Student feedback form	Screen	**[screen]** Menu > Operations > Support	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F396	Course feedback survey	Action	Student feedback form	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F397	Feedback analytics	Report	Student feedback form	Report card with chart/table + share pill	GL	Draw-in 600 ms
F398	Complaint escalation	Action	Student feedback form	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm
F399	Canned responses	Config	Student feedback form	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F400	Ticket SLA timers	Job	Student feedback form	No screen: status chip + notification	NE	Chip fade / pulse, no layout shift

M22 Security & Audit (20 functions)

Navigation: Drawer > System > Security **Roles:** Admin (owner) only



Materials: Glass rows, neon status edge, clay danger buttons.

Motion: Severity pulse, hold-to-confirm ring, list stagger.

1 Posture tile (F401) is the only screen that uses a strong neon edge: green = healthy, red = action needed.

2 Status rows (F407, F408, F417) are glass with severity chips.

3 Audit viewer (F402-F404) is read-only; filters are liquid chips; export requires re-auth.

4 Lock / unlock account (F409, F410) are clay danger actions with hold-to-confirm.

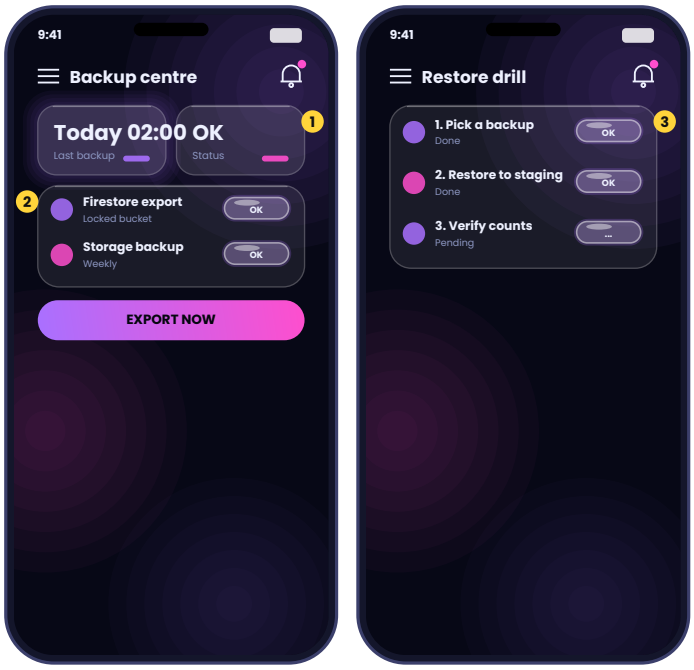
Figure 7.22 - Security & Audit: Security centre (left) and Audit log (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F401	Security centre	Screen	**[screen]** Menu > System > Security	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F402	Audit log viewer	Screen	**[screen]** Menu > System > Security	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F403	Audit filter and search	List	Audit log viewer	Glass search field, 300 ms debounce	GL	Field expand 250 ms, results fade-in
F404	Audit export	Report	Audit log viewer	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F405	Role and claim viewer	List	Audit log viewer	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F406	Suspicious sign-in alerts	Job	Audit log viewer	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F407	Device integrity status (Play Integrity)	List	Audit log viewer	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F408	App Check enforcement status	List	Audit log viewer	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F409	Lock account	Action	Audit log viewer	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F410	Unlock account	Action	Audit log viewer	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F411	Force password reset	Action	Audit log viewer	Security form sheet with re-authentication step	GL	Field-focus glow, error shake 420 ms
F412	Device / network allow rules (optional)	Config	Audit log viewer	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s

ID	Function	Type	Placed in	UI component	Mat	Animation
F413	Sensitive-data access log	List	Audit log viewer	Query / filter / list control inside host screen	LQ	Chip morph, list cross-fade 200 ms
F414	Consent and privacy requests	Screen	**[screen]** Menu > System > Security	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F415	Data erasure request handling	Action	Consent and privacy requests	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F416	Session policy config	Config	Consent and privacy requests	Session row with swipe-to-revoke	GL+CL	Swipe reveal red clay button, row collapse 200 ms
F417	Security-rules deployment status	List	Consent and privacy requests	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms
F418	Secret-rotation reminders	Job	Consent and privacy requests	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F419	Four-eyes approval for critical changes	Action	Consent and privacy requests	Approval card with Approve / Reject (money: re-auth)	CL	Hold-to-confirm ring, success tick + haptic
F420	Security incident log	Screen	**[screen]** Menu > System > Security	Security status row with severity chip	GL+NE	Severity chip pulse, row expand 250 ms

M23 Backup & Data Tools (14 functions)

Navigation: Drawer > System > Backup **Roles:** Admin (owner) only



Materials: Glass tiles/list, clay export button.

Motion: Progress ring, checklist tick draw.

1 Backup tiles (F421-F424) show last run and status; history is a glass list.

2 EXPORT NOW (F423) is a clay button that animates into a progress ring.

3 Restore drill (F425) is a checklist; each step draws its tick (200 ms) when verified.

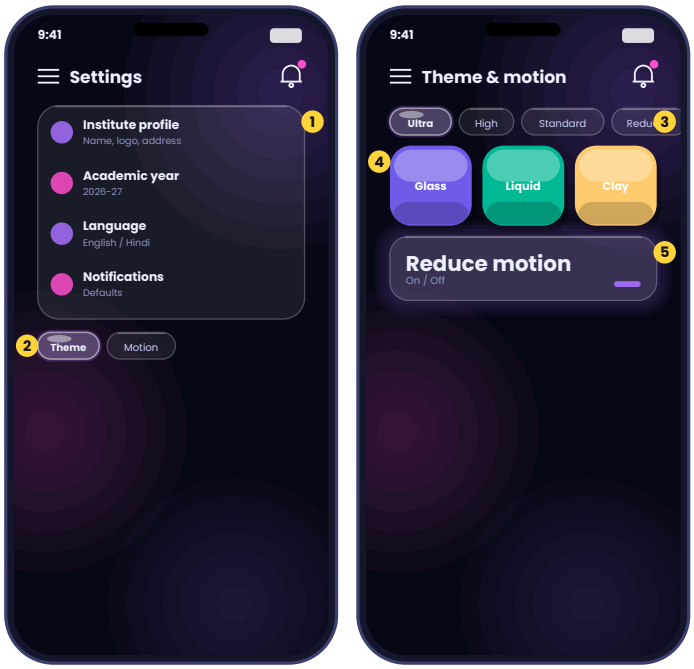
Figure 7.23 - Backup & Data Tools: Backup centre (left) and Restore drill (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F421	Backup centre	Screen	**[screen]** Menu > System > Backup	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F422	Scheduled Firestore export	Job	Backup centre	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F423	Manual export now	Action	Backup centre	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F424	Backup status and history	List	Backup centre	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F425	Restore drill checklist	Screen	**[screen]** Menu > System > Backup	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F426	Point-in-time recovery information	List	Restore drill checklist	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F427	Storage backup	Job	Restore drill checklist	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F428	Data integrity checker	Job	Restore drill checklist	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift
F429	Orphan record cleaner	Job	Restore drill checklist	Background job - no screen; surfaces as status chip / notification	NE	Status-chip pulse, no layout shift

ID	Function	Type	Placed in	UI component	Mat	Animation
F430	Import validator	Action	Restore drill checklist	File picker + validation-result sheet	LQ	Progress bar, row-count ticker
F431	Data retention rules	Config	Restore drill checklist	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F432	Archive old academic years	Action	Restore drill checklist	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring
F433	Export CSV bundle	Report	Restore drill checklist	Export / share pill + system share sheet	LQ	Pill morphs to progress 400 ms
F434	Delete-protection status	List	Restore drill checklist	Backup status card + checklist	GL	Checklist tick draw 200 ms, progress ring

M24 Settings & Institute Profile (16 functions)

Navigation: Drawer > System > Settings **Roles:** Admin (owner)



Materials: Glass sections, liquid tier chips, clay material preview tiles.

Motion: Preview cross-fade, switch spring.

1 Settings list (F435-F442) uses grouped glass sections with 1 dp dividers.

2 Theme intensity (F444) and reduce motion (F445) are linked: Reduced tier removes parallax and shaders.

3 Live preview tiles show Glass, Liquid and Clay so the owner can see the effect before saving.

4 Feature flags (F443) live in a secondary section with remote-config source labels.

5 Institute profile (F436) edits logo and receipt header (F437).

Figure 7.24 - Settings & Institute Profile: Settings (left) and Theme & motion (right).

ID	Function	Type	Placed in	UI component	Mat	Animation
F435	Settings home	Screen	**[screen]** Menu > System > Settings	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F436	Institute profile (name, logo, address)	Screen	**[screen]** Menu > System > Settings	Media picker bottom sheet + progress ring	LQ	Ring progress, thumbnail scale-in 250 ms
F437	Branding colours and receipt header	Config	Institute profile (name, logo, address)	Document preview card + share pill	GL+LQ	Fade-through 300 ms, share-sheet slide-up
F438	Academic year settings	Config	Institute profile (name, logo, address)	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F439	Working days and hours	Config	Institute profile (name, logo, address)	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F440	Language and locale	Config	Institute profile (name, logo, address)	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F441	Currency and tax settings	Config	Institute profile (name, logo, address)	Settings row (switch / slider / picker)	GL	Thumb spring, saved toast 1.8 s
F442	Notification defaults	Config	Institute profile (name, logo, address)	Banner / inbox row / push composer	GL	Banner slide-down 250 ms
F443	Feature flags (Remote Config)	Config	Institute profile (name, logo, address)	Settings switch / slider row	GL	Thumb spring, haptic confirm
F444	Theme: glass / liquid intensity	Config	Institute profile (name, logo, address)	Settings switch / slider row	GL	Thumb spring, haptic confirm
F445	Reduce motion and accessibility	Config	Institute profile (name, logo, address)	Settings switch / slider row	GL	Thumb spring, haptic confirm
F446	Storage and cache settings	Config	Institute profile (name, logo, address)	Settings switch / slider row	GL	Thumb spring, haptic confirm
F447	App version and update prompt	List	Institute profile (name, logo, address)	AI chat / admin AI panel	GL	Streaming text reveal, typing dots
F448	Legal pages (privacy / terms)	Screen	**[screen]** Menu > System > Settings	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms

ID	Function	Type	Placed in	UI component	Mat	Animation
F449	About and diagnostics	Screen	**[screen]** Menu > System > Settings	Full destination: Scaffold + glass top bar	GL	Shared-axis X transition 300 ms
F450	Admin profile edit	Action	About and diagnostics	Button / bottom sheet / dialog inside host screen	CL+LQ	Press scale .96 spring, haptic on confirm

8. Teacher, Student and Parent Portals

These portals reuse the same functions as the admin portal with narrower scopes (architecture chapter 11). The same components and materials are used, but the portal has its own accent colours, a lighter chrome and fewer destinations.

8.1 Teacher portal

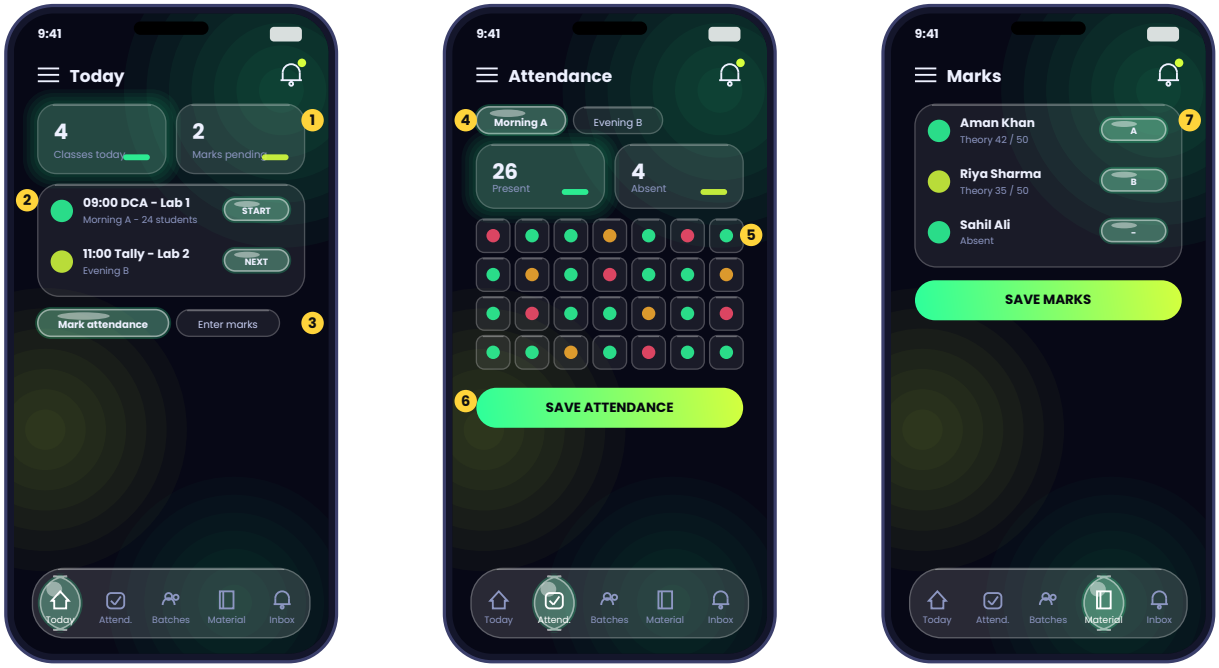


Figure 8.1 - Teacher portal: Today, Attendance, Marks.

1 Today (F019-style tiles, F111 timetable, F125) is the home of the teacher: next class card with a START button that opens the live class (F317).
2 Quick chips jump to attendance (F126-F130) and marks (F202) for the current batch.
3 Large 48 dp roster cells; haptic tick per mark; bulk mark all present (F130).
4 Save works offline; WorkManager queues attendance (F144) and a "queued" chip shows.
5 Marks grid (F202) shows only own batches; grade chip updates as marks are typed.
6 Teacher sees no fee, payroll, security or backup screens; those routes do not exist in the teacher graph.

8.2 Student portal



Figure 8.2 - Student portal: Home, Timetable, AI tutor.

1 Home uses clay tiles for the three most used areas (Fees F164, Results F205-F210, AI tutor F336) so the screen feels friendly and tactile.
2 Next class and assignment cards are glass; Join live (F318) appears 10 minutes before class.
3 Timetable (F111) is read-only; the current slot glows with a neon edge.
4 AI tutor (F336) shows a quota bar (F341) and streams the answer.
5 Student can download material (F215-F220), see certificates (F233-F246) and open support (F389).

8.3 Parent portal

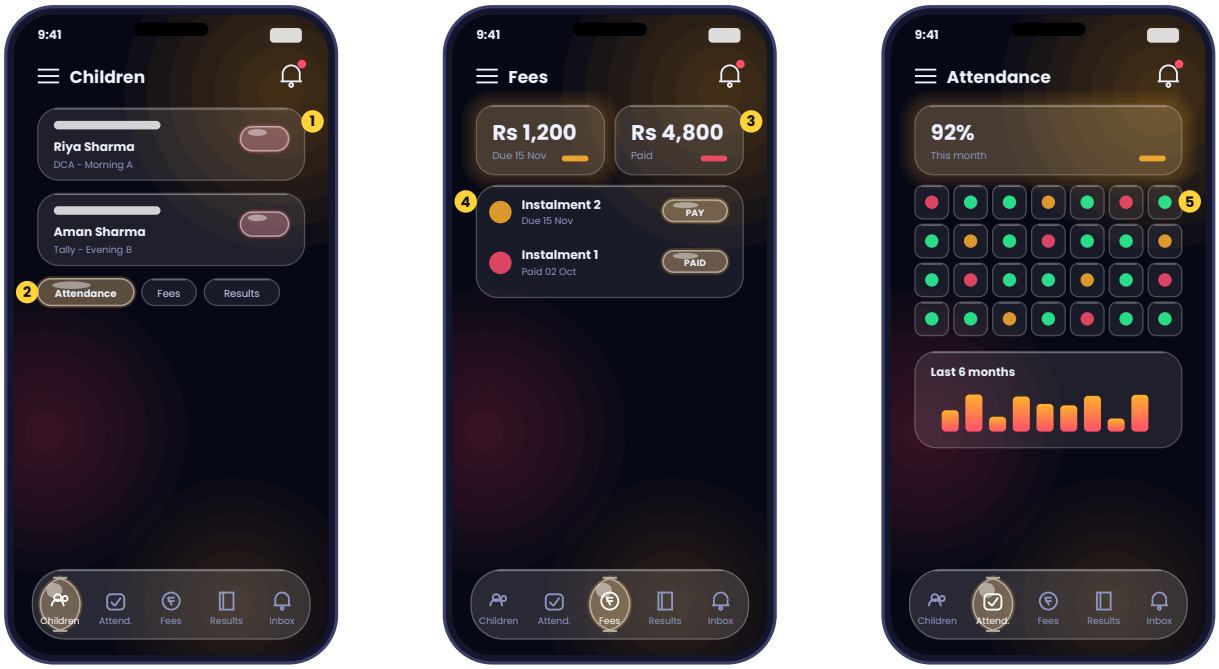


Figure 8.3 - Parent portal: Children, Fees, Attendance.

1 Children tab (F056, F288) lists linked children with status chips; switching child crossfades the whole portal.
2 Quick chips jump to the child attendance, fees and results; everything is read-only.
3 Fee tiles (F164) show due and paid; online payment (F151) opens the gateway through a server-verified flow.
4 Receipts (F153) are downloadable PDFs.
5 Attendance calendar (F134) uses colour plus icon so colour-blind parents can read it.

8.4 Module visibility by role

Module	Admin	Teacher	Student	Parent
M01 Authentication	Full	Own	Own	Own
M02 Dashboard	Full	Today view	-	-
M03 Students	Write	Read own batches	Self	Linked children
M04 Admissions	Full	-	-	-
M05 Courses	Write	Read	Read (enrolled)	-
M06 Batches & timetable	Write	Own batches	Own	Linked children
M07 Attendance	Full	Own batches	Self	Linked children
M08 Fees	Write	-	Self	Linked children
M09 Accounts	Full	-	-	-
M10 Exams & results	Full	Own batches	Self	Linked children
M11 Material & assignments	Full	Write	Read assigned	-
M12 Certificates	Full	-	Own	-
M13 Staff	Full	-	-	-
M14 Payroll & leave	Full	Own payslip, leave	-	-
M15 Guardians	Full	-	-	Own profile
M16 Inbox	Send all	Send to own batches	Receive	Receive
M17 Live classes	Full	Host	Join assigned	-
M18 AI hub	Assistant + config	Tutor	Tutor (quota)	-
M19 Reports	Full	Own batches, limited	-	-
M20 Inventory	Full	-	-	-
M21 Support	Full	Create / read	Create / read own	Create / read own
M22-M24 Security, Backup, Settings	Owner only	-	-	-

The UI hides what a role cannot use, and Firestore Rules or Cloud Functions refuse it regardless. A hidden screen is never the only protection.

9. Animation and Motion Catalogue

Thirty-two named animations cover the whole app. Each has an ID so a developer or designer can refer to it ("use A13 on this tile"). Every animation has a reduced-motion fallback. All durations come from the motion tokens in Chapter 3.

ID	Name	Used by	Specification	Compose	Reduced
A01	Launch splash to login	F001	Icon scale + fade 300 ms, then login entrance	SplashScreen API + AnimatedVisibility	Instant
A02	Login card entrance	F001	Rise 40 dp, rotateX 18 deg to 0, scale .96 to 1; 1000 ms, easeStandard	Animatable + graphicsLayer	Fade 200 ms
A03	Conic border sweep	F001	One turn every 5 s, linear loop	Modifier.neonBorder	Off
A04	Role pill spring	F001	Spring damping .55, stiffness 300; accent colour crossfade 700 ms	animateFloatAsState + animateColorAsState	Crossfade
A05	Role content swap	F001	Fade + rise 14 dp + scale .98 + blur 6 to 0; 550 ms	AnimatedContent (+ Modifier.blur on API 31+)	Fade 150 ms
A06	CTA shimmer	F001, F043, F149, all primary CTAs	Light sweep every 3.2 s while enabled	drawWithContent + infinite transition	Off
A07	Verify spinner	F002	700 ms per turn; button disabled	CircularProgressIndicator	Same
A08	Error shake	F003, F007, F131	Offset -9, 9, -5, 0 over 420 ms; red flash	Animatable keyframes	Colour flash only
A09	Success pulse rings	F002	Three expanding rings 1.5 s; wave amplitude boost	Canvas + shader uniform	Tick only
A10	Gyro tilt	F001, F019	Up to 7 deg following device tilt, low-pass filtered	SensorManager + graphicsLayer rotation	Off
A11	Tab bar pill	All tabs	Bouncy spring; icon scale 1.0 to 1.15; label fade	animateFloatAsState	Jump
A12	Drawer open	Drawer items (F019 area)	Content scales to .94 and dims; items stagger 30 ms	ModalNavigationDrawer + graphicsLayer	Fade
A13	KPI count-up	F020-F026, F165	0 to value in 600 ms, ease out	Animatable / animateIntAsState	Final value
A14	Money ticker	F023, F149, F164	Digit roll 500 ms with tabular numerals	AnimatedContent per digit	Instant
A15	Chart draw-in	F031, F032, F097, F141, F142, F184	Path trim 700 ms; tooltip spring	Vico animation / Canvas Path	Static
A16	List stagger	F035, F155, F303	40 ms stagger, 250 ms each (first 8 rows only)	Modifier.animateItem + delay	None
A17	Skeleton shimmer	Paged lists, F048	1.2 s sweep on grey glass blocks	Brush shimmer	Static grey
A18	Pull to refresh	F033	Liquid drop stretches then snaps; spring	PullToRefreshBox + custom indicator	Spinner
A19	Swipe actions	F012, F304	Row slides to reveal action; snap 200 ms	SwipeToDismissBox / AnchoredDraggable	Overflow buttons
A20	Roster cell flip	F126-F130	rotationY 180 ms + haptic tick	graphicsLayer	Colour swap
A21	Glass bottom sheet	F038-F041, F299, F112	Slide up 400 ms springGentle; scrim 40%	ModalBottomSheet with glass container	Fade
A22	Container transform	F035 to F048, F265	Card grows into detail 400 ms	SharedTransitionLayout	Fade-through
A23	Kanban drag	F070-F073	Lift (elevation up, 3 deg tilt); settle spring on drop	detectDragGesturesAfterLongPress	No tilt
A24	Switch spring	F352, F443-F445	Thumb springSnappy; haptic confirm	Custom Switch	Instant
A25	Success tick + confetti	F149, F206, F236	Lottie tick 900 ms; confetti 1.2 s (big events only)	Lottie Compose + Konfetti	Tick only
A26	Hold-to-confirm ring	F161, F310, F409, F052	Ring fills 800 ms; release cancels	Animatable + pointerInput	Tap + dialog
A27	Scan line	F127, F387	1.6 s sweep over viewfinder; success pulse	Infinite transition	Static frame
A28	Streaming text	F336-F339	Chunks appear every ~30 ms with soft fade	State append + AnimatedContent	Instant
A29	Call controls auto-hide	F321-F326	Fade after 3 s idle; tap shows 250 ms	AnimatedVisibility	Always visible
A30	3D hero float	Empty states, AI hub header (Ultra tier)	6 s sine float +/- 6 dp; gyro parallax	Filament / SceneView, Lottie fallback	Static image
A31	Banner slide	F069, F301, F310, F018	Slide down 250 ms, auto-dismiss 4 s	AnimatedVisibility	Fade
A32	Checklist tick draw	F425, F259, F260	Stroke draws in 200 ms	Canvas path trim	Instant

9.1 Haptics map

Moment	Haptic	Notes
Roster mark, chip select, tab change	Light tick	Debounced to one per 50 ms

Moment	Haptic	Notes
Save success, payment posted, result published	Confirm (where available) else long-press	Paired with A25
Validation or wrong password	Reject (where available) else double tick	Paired with A08
Hold-to-confirm completes	Heavy click	Paired with A26
Scrolling, typing, animations	None	Haptics are never continuous

9.2 Reduced-motion behaviour

The Reduced tier is chosen when the system animation scale is 0, battery saver is on, or the owner enables "Reduce motion" (F445). In this tier: no parallax or tilt, no shaders or 3D, no infinite loops (border sweep, shimmer, wave), springs become 150 ms fades, and confetti is skipped. Information is never carried by motion alone.

9.3 Three useful snippets

```
// A13 count-up
@Composable fun CountUp(target: Int, style: TextStyle) {
    val v = remember { Animatable(0f) }
    LaunchedEffect(target) { v.animateTo(target.toFloat(), tween(600, easing = FastOutSlowInEasing)) }
    Text(NumberFormat.getIntegerInstance(Locale("en", "IN")).format(v.value.toInt()), style = style.copy(fontFeatureSettings
}

// A10 gyro tilt (cap 7 degrees, low-pass filtered, off on Reduced tier)
fun Modifier.tiltWithGyro(maxDeg: Float = 7f) = composed {
    val tier = LocalUiTier.current; var rx by remember { mutableFloatStateOf(0f) }; var ry by remember { mutableFloatStateOf(
    if (tier != UiTier.Reduced) DisposableEffect(Unit) { /* register TYPE_GAME_ROTATION_VECTOR, smooth, clamp to +/- maxDeg *
    graphicsLayer { rotationX = rx; rotationY = ry; cameraDistance = 12f * density }
}

// A22 container transform
SharedTransitionLayout { AnimatedContent(selectedId) { id ->
    if (id == null) StudentList(Modifier.sharedBounds(rememberSharedContentState("card"), this@AnimatedContent))
    else StudentDetail(id, Modifier.sharedBounds(rememberSharedContentState("card"), this@AnimatedContent)) } }
```

10. Icons, Illustrations, 3D and Empty States

Asset type	Format and limits	Rule
Icons	Vector drawables (Material Symbols, rounded, 1.8 dp stroke look); role icons as in the login reference (shield, cap, board, family)	Never import the full extended icon set; ship only the used icons.
Illustrations / empty states	Lottie JSON under 100 KB each, or static WebP	One illustration per empty state; colours follow the role accent.
3D hero	glTF / GLB under 1.5 MB, compressed textures; rendered by Filament / SceneView on Ultra tier only	Pause off-screen; fallback is a pre-rendered WebP.
Photos	Coil with memory and disk caps; thumbnails 256 px, full 1080 px	Student photo is circle-cropped; ID proofs open in a zoomable viewer.
Fonts	One variable sans family bundled as resources	Tabular numerals enabled for money and counters.
App icon	Adaptive icon + monochrome (themed) layer; splash with the same mark	Colours match the Admin pair.

10.1 Empty-state copy (real empty states, never sample rows)

Screen	Illustration	Text	Action
Students list	Clay group of people	No students yet. Add the first student.	ADD STUDENT (F043)
Fees	Clay wallet	No payments recorded today.	TAKE FEES (F149)
Attendance	Clay calendar tick	No batch scheduled today.	OPEN TIMETABLE (F111)
Inbox	Bell	You are all caught up.	-
Exams	Clay paper	No exams created yet.	CREATE EXAM (F192)
Live classes	Camera	No live class scheduled.	SCHEDULE (F316)
AI tutor	3D spark (Ultra) / Lottie	Ask your first question.	Composer focus
Reports	Chart	Reports appear once data is entered.	-
Audit log	Shield	No events match this filter.	CLEAR FILTER
Parent: children	Family	No child is linked yet. Ask the institute to link your number.	CONTACT SUPPORT (F390)

11. Libraries for the UI Layer

The architecture blueprint (Appendix B) already selected 50 libraries. The table lists the ones that shape the UI, then the **additional UI libraries proposed in this document**. Versions are not frozen here; each is checked against its release notes and maintenance status when it is added, and risky ones are wrapped behind an interface.

11.1 Already approved UI libraries (Appendix B numbers)

#	Library	Role in this design
1-5	Compose BOM + UI, Material 3, Foundation + Animation, Navigation Compose, Lifecycle ViewModel	Base toolkit, transitions, state
6	Haze	Backdrop blur for GlassCard on High and Ultra tiers
7	Backdrop / liquid-glass library	Lens and refraction on Ultra; wrapped by LiquidSurface
8	AGSL RuntimeShader (platform)	Wave background, shader ripples (API 33+)
9	Filament / SceneView	3D hero scenes (Ultra only)
10	Lottie Compose	Success tick, empty states, loaders
11	Coil 3	Photos and thumbnails
12	Vico	All charts
13-15	Media3 ExoPlayer, CameraX, ML Kit Barcode	Recorded lectures, photo capture, QR scan
16	Splash Screen API	A01

11.2 New libraries added for the UI

Library	Purpose	Used by	Notes / risk
androidx.graphics:graphics-shapes	Shape morphing for liquid blobs, pill-to-circle transitions	A04, A11, FAB	AndroidX, low risk
Konfetti (Compose)	Confetti burst on big successes	A25	Small; used only for result publish and certificate issue
Kizitonwose Calendar (Compose)	Month and week calendars with custom day cells	F111, F119, parent attendance calendar	Well maintained; wrap day cell in design system
Reorderable (Compose)	Drag-and-drop reorder lists	F034, F090	Fallback: up/down buttons
Telephoto (zoomable image)	Pinch zoom for ID proofs and certificates	F058, F243	Pairs with Coil
Material Kolor (optional)	Build a palette from a seed colour for institute branding	F437	Optional; fixed role palettes remain default
Compose Material3 Adaptive + window size class	Navigation rail, list-detail two-pane on tablets	Shell on width 600 dp+	AndroidX
androidx.core:core-performance	Device performance class input to tier detection	Tier probe	Optional; tier also uses RAM and API level
androidx.profileinstaller	Installs Baseline Profile for fast start and smooth lists	Startup budget	AndroidX
Roborazzi (or Paparazzi)	JVM screenshot tests for Standard / Reduced tiers and light/dark	CI	JVM tools do not render real blur; Ultra and High tiers verified on devices
Compose Shimmer (optional)	Skeleton loading	A17	We ship our own brush; adopt only if it saves code

11.3 Considered and not adopted

- **Rive**: powerful but adds runtime size; Lottie is enough for v1.
- **Orbital**: replaced by Compose built-in SharedTransitionLayout.
- **Full extended Material icons**: too large; only used icons are bundled.
- **Cloudy (blur for older Android)**: not needed, because Standard tier deliberately has no blur; reconsider only if owners want blur on very old phones.
- **Any WebView-based chart or UI library**: forbidden by rule U1.

12. Accessibility, Performance and UI Testing

12.1 Accessibility

Area	Requirement	How glass and motion comply
Contrast	Text at least 4.5:1, large text 3:1, icons 3:1	Add 8% dark scrim behind text on bright backdrops; contrast checked in screenshot tests on the brightest orb positions.
Touch targets	48 x 48 dp minimum	Roster cells, chips and icon buttons padded to 48 dp.
Screen reader	Every control has a content description and role; roster cells announce "Present, Aman Khan"	Decorative orbs, waves and glows are hidden from TalkBack.
Font scale	Up to 200%	Cards grow vertically; no fixed heights on text containers; tiles wrap labels.
Colour independence	Status is not colour-only	Attendance and dues cells carry an icon or letter (P, A, L).
Motion	Respect system and in-app reduce-motion	Reduced tier (9.2); nothing flashes more than 3 times per second.
Focus and keyboard	Visible focus ring; logical order	Neon ring doubles as focus indicator.
Language	English and Hindi strings	All text in resources; layouts tolerate longer Hindi text.

12.2 Performance budgets for the UI

Metric	Budget	Verified by
Cold start to interactive login	Under 2 s on a mid-range phone	Macrobenchmark + Baseline Profile
Frame time on lists	No dropped frames at 60 Hz on Standard tier; Ultra within 90 percent	Macrobenchmark frame timing
Live blur layers	Maximum 2 per screen	Design review checklist + code review
Wave background	Under 2 ms GPU per frame on a mid-range phone; paused when not visible	GPU profiler on device matrix
Overdraw	Glass areas not stacked more than 2 deep	GPU overdraw tool
APK size impact of UI assets	Tracked per release; Lottie + 3D under 6 MB total	CI size report

12.3 Testing the visuals

Layer	What	Tool
Component gallery (debug builds only)	Every material and component in all four tiers and both themes. Contains no student data and is excluded from release builds.	Debug source set
Screenshot tests	Key screens x 4 portals x light/dark x font scale 1.0 and 2.0, using Standard and Reduced tiers	Roborazzi / Paparazzi
Device tests	Ultra and High tiers on real devices (blur, shaders, 3D)	Manual + Firebase Test Lab if budget allows
Accessibility checks	Automated checks plus a TalkBack pass	Compose UI Test + Accessibility Test Framework
Motion review	Each A-ID recorded once and compared to the spec	Screen recording on a reference device

13. Delivery Plan, CI Guards and Acceptance

13.1 UI work by roadmap phase (matches architecture chapter 18)

Phase	UI deliverable	Modules / chapters
0 Foundation	Tokens, theme, tier detection, GlassCard, LiquidPill, ClayButton, neon modifiers, debug component gallery, Neon Wave login, admin shell	Ch. 3-6, M01
1 Core operations	Dashboard, Students, Attendance, Fees, Inbox, batches and timetable; teacher Today + attendance	M02, M03, M05-M08, M16
2 Admissions and finance	Pipeline board, admission form, accounts, payroll, guardians; parent Children + Fees	M04, M09, M14, M15
3 Academics	Exams, marks grid, material, certificates, reports; student Home, Results, Timetable	M10-M12, M19
4 Live and AI	Call controls, AI chat, usage dashboard; 3D hero (Ultra)	M17, M18
5 Operations and hardening	Staff, inventory, support, security, backup, settings; accessibility pass; Baseline Profile tuning	M13, M20-M24
6 Launch	Screenshot baselines locked; motion review; store assets	All

13.2 CI guards for "native only"

These checks extend the guard step already described in the architecture blueprint (chapter 16.2). They make the "no web app files" rule enforceable by the build, not by memory.

```
# add to .github/workflows/android-ci.yml (verify job)
- name: Guard - native UI only
  run: |
    if grep -R --include=*.kt --include=*.java --include=*.xml -nE "android\.webkit\.WebView|<WebView|WebViewClient|loadUrl\(
      echo "WebView usage is forbidden"; exit 1; fi
    if find app/src -path "*assets*" \( -name "*.html" -o -name "*.js" -o -name "*.css" \) | grep -q .; then
      echo "Web assets are forbidden"; exit 1; fi
    if grep -R -nE "three(\.min)?\.js|cdnjs\.cloudflare|cdn\.jsdelivr" app core feature; then
      echo "Web script reference found"; exit 1; fi
    if grep -R --include=*.kt -nE "Color\(0x[0-9A-Fa-f]{8}\)" feature; then
      echo "Hard-coded colours in feature modules; use theme tokens"; exit 1; fi
```

13.3 Screen acceptance checklist (per screen)

#	Check	Evidence
1	Every function listed for the screen in Chapter 7 / 8 is reachable and works against the real project	Test log
2	Uses only theme tokens (colour, type, shape, motion)	Lint report
3	Material policy respected (2.3); at most two live blur layers	Code review
4	Loading, empty, error, offline and permission-denied states designed (no sample rows)	Screenshots
5	Animations match their A-IDs and have reduced-motion fallbacks	Screen recording
6	Contrast, touch target, TalkBack and 200% font checked	A11y checklist
7	Works on Standard and Reduced tiers; Ultra verified on a device	Device log
8	Hindi and English strings present; no text clipped	Screenshots
9	Frame time inside budget on the list or main interaction	Macrobenchmark
10	No WebView, no web asset, no hard-coded secret	CI guards

13.4 Open decisions for the owner

- Final institute logo and wordmark for the login and app icon (the attached reference carries another brand).
- Whether Parent login is phone OTP only, or OTP plus password (affects the Parent tab, Figure 5.3).
- English only at launch, or English + Hindi strings (architecture decision 19.2).
- Preferred default tier on the institute devices (Ultra, High or Standard) and whether 3D heroes are wanted at all.
- Light theme at launch, or dark only first (dark is the primary design in this document).

Final verdict. The design is complete enough to start Phase 0: tokens, materials and the Neon Wave login can be built and reviewed on real phones before any module screen is coded. Every statement here remains a proposal until it is verified on devices and in the real repository.

Appendix A. Function Coverage Index

Every function from the architecture blueprint appears in exactly one module table in Chapter 7. This index counts them by type, so nothing is missed.

Module	Name	Screen	Action	List	Report	Job	Config	Total
M01	Authentication & Session	2	12	1	0	3	0	18
M02	Admin Dashboard	1	2	10	2	0	1	16
M03	Student Management	4	17	8	1	0	0	30
M04	Admissions & Enquiries	4	11	0	2	2	1	20
M05	Course Catalog	3	7	1	1	0	6	18
M06	Batch & Timetable	3	14	3	1	1	0	22
M07	Attendance	1	8	2	3	6	0	20
M08	Fees & Payments	2	13	3	6	2	2	28
M09	Accounts & Expenses	3	9	0	4	0	2	18
M10	Examinations & Results	4	11	0	4	3	2	24
M11	Assignments & Study Material	2	8	3	1	1	3	18
M12	Certificates	2	8	0	1	3	0	14
M13	Teacher & Staff Management	4	14	3	1	0	0	22
M14	Payroll & Leave	1	9	1	2	0	3	16
M15	Guardian (Parent) Linking	2	6	0	0	0	4	12
M16	Notifications & Announcements	2	8	2	1	2	3	18
M17	Live Classes (Agora)	2	11	1	2	4	0	20
M18	AI Tutor & Assistant	3	5	1	1	5	3	18
M19	Reports & Analytics	2	4	1	13	1	1	22
M20	Inventory, Lab & Library	3	7	0	1	2	1	14
M21	Support & Feedback	3	6	0	1	1	1	12
M22	Security & Audit	4	5	6	1	2	2	20
M23	Backup & Data Tools	2	3	3	1	4	1	14
M24	Settings & Institute Profile	4	1	1	0	0	10	16
	Total	**63**	**199**	**50**	**50**	**42**	**46**	**450**

Total: 450 functions.